

Metrics of MT Quality

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unless otherwise stated

Course Outline

1. **Metrics of MT Quality.**
2. Approaches to MT. SMT, PBMT, NMT, NP-hardness.
3. NMT (Seq2seq, Attention. Transformer).
4. Parallel texts. Sentence and word alignment. hunalign, GIZA++.
5. PBMT: Phrase Extraction, Decoding, MERT. Moses.
6. Morphology in MT. Factors or segmenting, data or linguistics.
7. Syntax in SMT (constituency, dependency, deep).
8. Transformer (and syntax in NMT). LLM-Based Translation.
9. Towards Understanding: Word and Sentence Representations.
10. Multi-Lingual MT, Transfer Learning.
11. Multi-Modal Translation.
12. **Project presentations.**

Outline

- Task of MT (formulating a simplified goal).
- Evaluation as key component of research progress.
- Manual evaluation.
- Automatic evaluation.
- “LLMs as Judges”.
- Empirical confidence bounds.
- End-to-end vs. component evaluation.
- Summary: Evaluation caveats.

Our Goal in MT

We restrict the task of MT to the following conditions.

- No writers' ambitions, we prefer literal translation.
- No attempt at handling cultural differences.

Expected output quality:

1. Worth reading. (Not speaking the src. lang. I can sort of understand.)
2. Worth editing. (I can edit the MT output to obtain publishable text.)
3. Worth publishing, no editing needed.

In general, we're aiming at level 1 or 2. Level 3 remains risky.

Evaluation as Key Part of Research

Preparation

Translation

Evaluation

Training datasets

Baseline systems

Evaluation as Key Part of Research

Preparation

Translation

Evaluation

Training datasets

Baseline systems

New source texts

Evaluation as Key Part of Research

Preparation

Translation

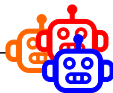
Evaluation

Training datasets

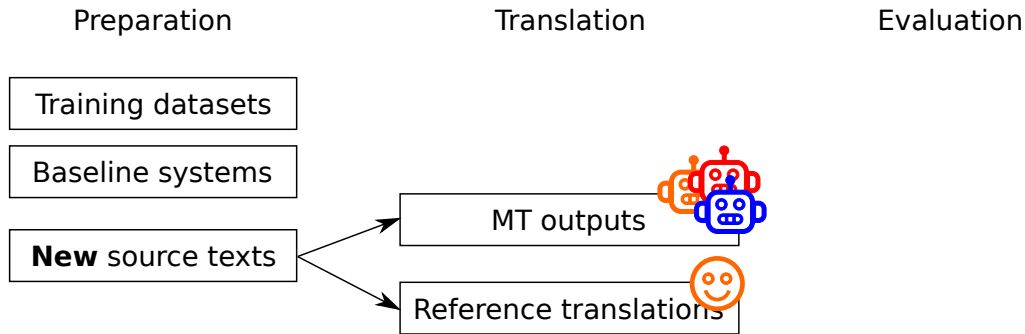
Baseline systems

New source texts

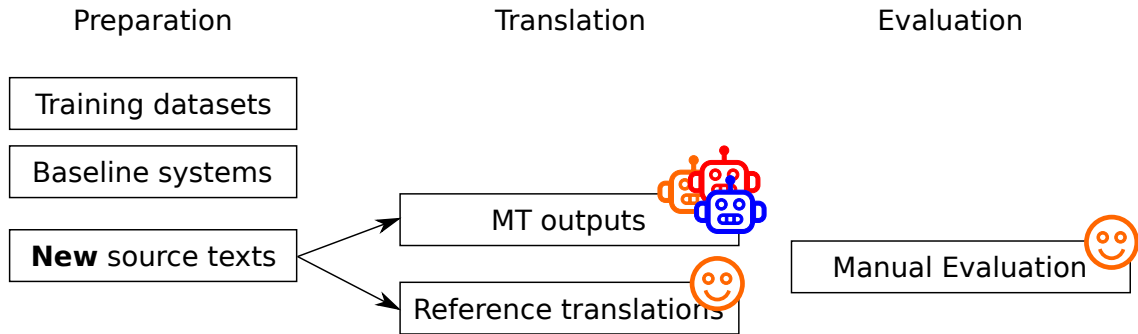
MT outputs



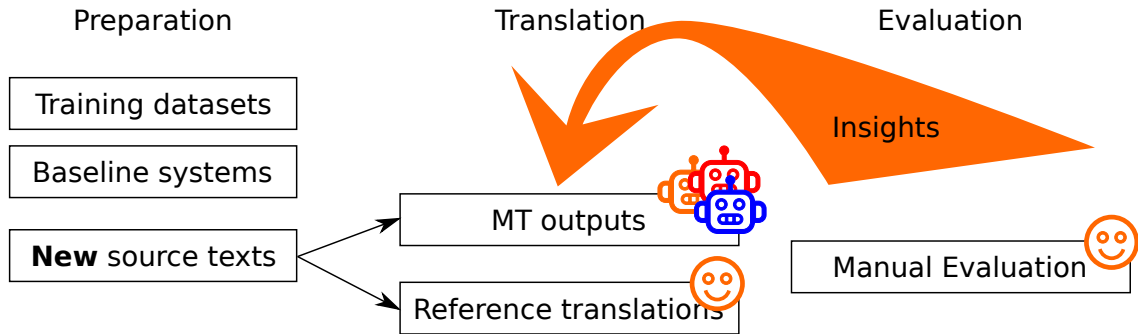
Evaluation as Key Part of Research



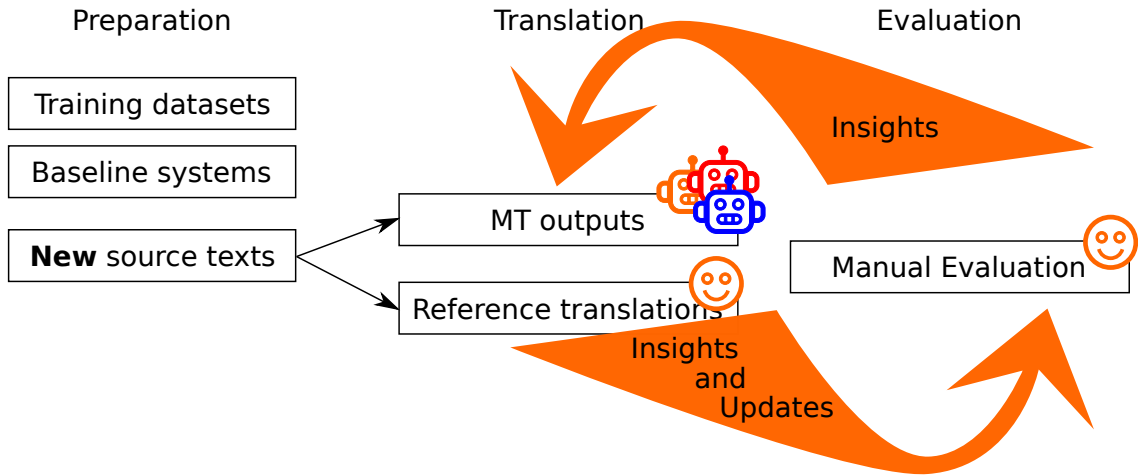
Evaluation as Key Part of Research



Evaluation as Key Part of Research



Evaluation as Key Part of Research



Annual “Competitions” in Translation (WMT)

Language Pair	'06	'07	'08	'09	'10	'11	'12	'13	'14	'15	'16	'17	'18	'19	'20	'21	'22	'23	'24
Chinese ↔ English												●	●	●	●	●	●	●	●
Czech ↔ English		●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Czech ↔ Ukrainian																	●	●	●
Finnish ↔ English										●	●	●	●	●					
French ↔ English	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
German ↔ English	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
German ↔ French														●	●	●	●	●	●
Gujarati ↔ English														●					
Hindi ↔ English									●										●
Icelandic ↔ English																●			●
Japanese ↔ Chinese																			●
Japanese ↔ English															●	●	●	●	●
Lithuanian ↔ English														●			●		
Russian ↔ English								●	●	●	●	●	●	●	●	●	●	●	●
Spanish ↔ English	●	●	●	●	●	●	●	●											●
Turkish ↔ English											●	●	●	●					
Ukrainian ↔ English																	●	●	●
...																			

Language pairs in the main translation task. <http://www.statmt.org/wmt24/>

●/● denote translation only in one direction (mostly out of English).

Dashed vertical lines indicate technological breakthroughs.

Importance of Measuring MT Output

You need a metric to be able to check your progress.

An example from the history:

- Manual judgement at Euratom (Ispra) of a Systran system (Russian→English) in 1972 revealed huge differences in judging; (Blanchon et al., 2004):
 - 1/5 (D-) for output quality (evaluated by teachers of language),
 - 4.5/5 (A+) for usability (evaluated by nuclear physicists).

Metrics can drive the research for the topics they evaluate.

- Some measured improvement required by sponsors: NIST MT Eval, DARPA, TC-STAR, EuroMatrix+.
- BLEU has lead to a focus on phrase-based MT.
- Other metrics may similarly change the community's focus.

Basic Manual Evaluation Decisions

What to Show to the annotators when assessing the candidate?

- **REF-based** ... only the (human) reference
- **SRC-based** ... only the source
- **SRC&REF-based** ... both

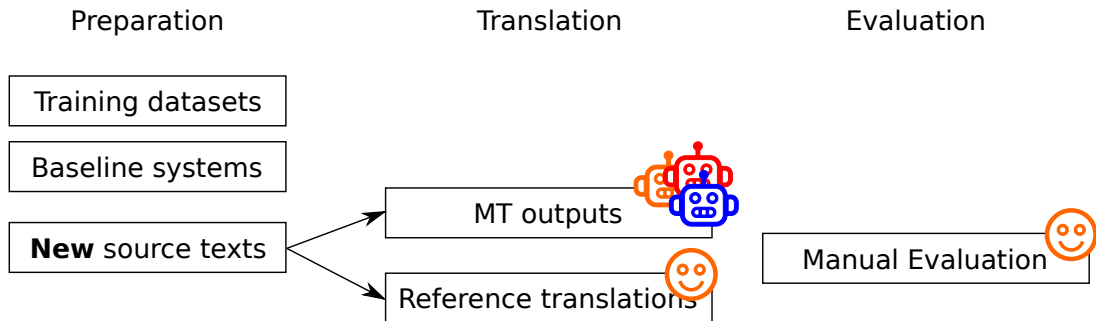
Context to Consider:

- **Sentence-level** ... sentences in random order
- **Document-level** ... obtain single score per document
- **Document-aware** ... show whole documents, scores per sentence

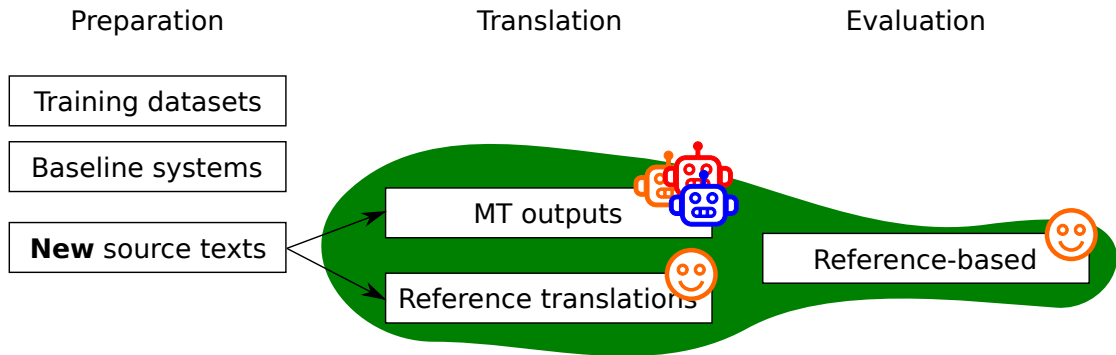
What to Ask from annotators (scoring technique):

- Some **relative** score over several candidates?
- Some **absolute** score for a single output?
- A **more complicated question**?

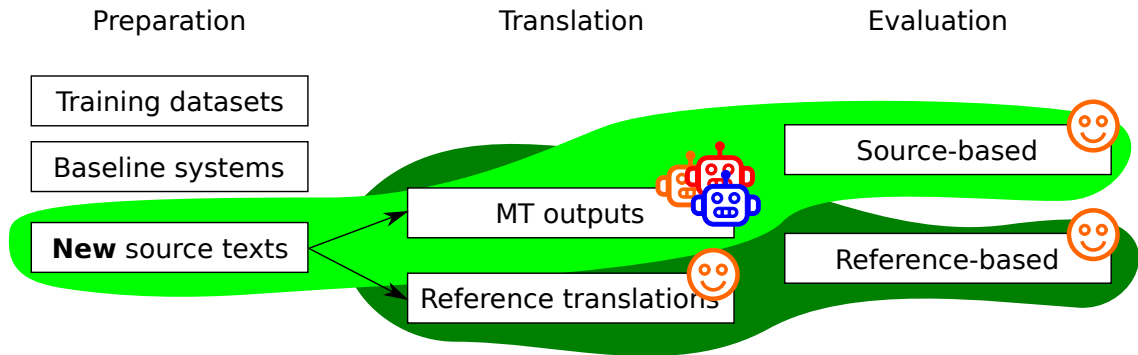
Source vs. Ref-Based Evaluation



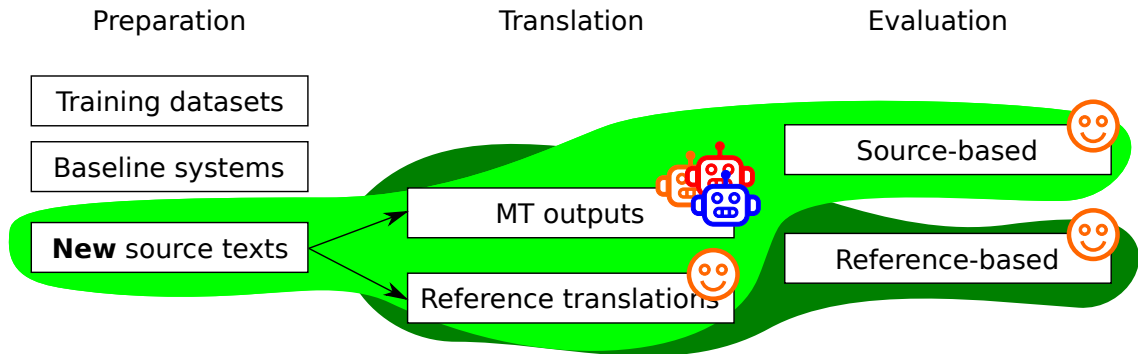
Source vs. Ref-Based Evaluation



Source vs. Ref-Based Evaluation



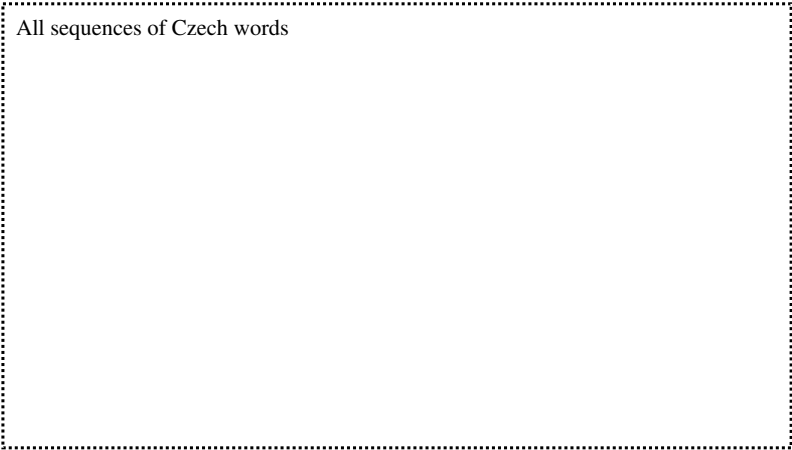
Source vs. Ref-Based Evaluation



Space of Possible Translations

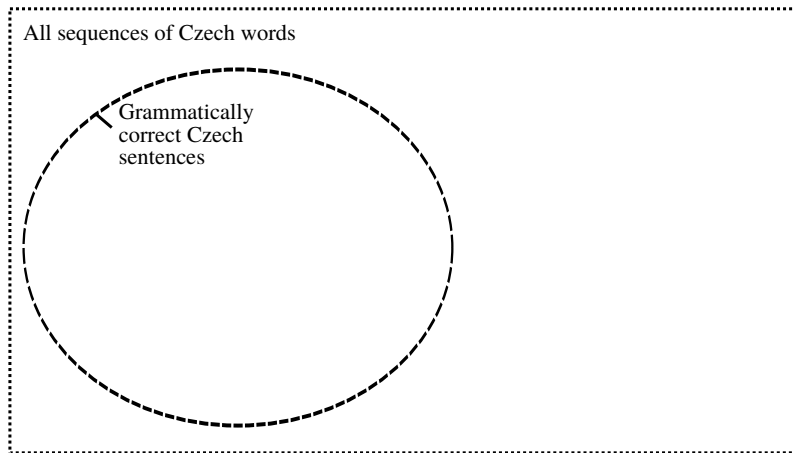
For a single English source sentence, we can consider:

All sequences of Czech words



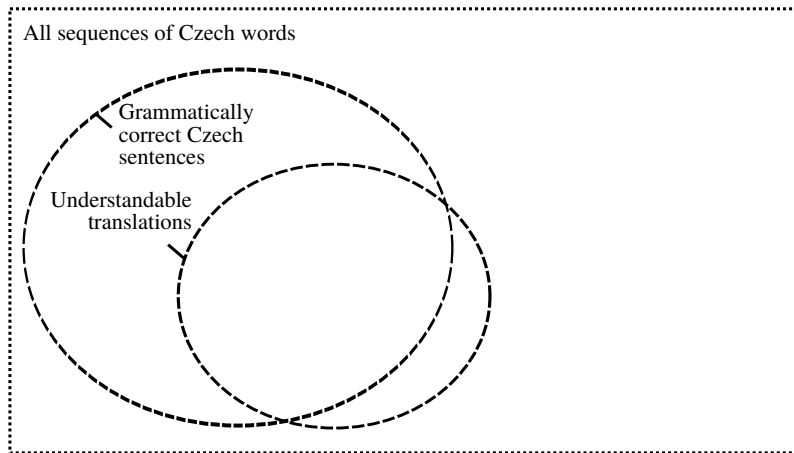
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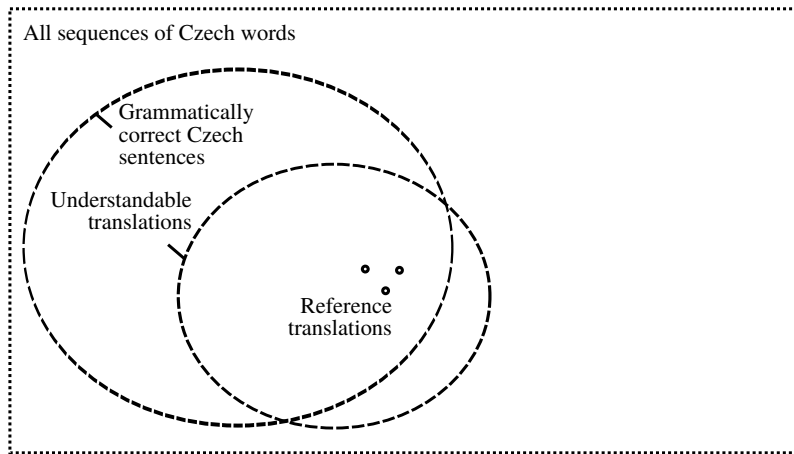
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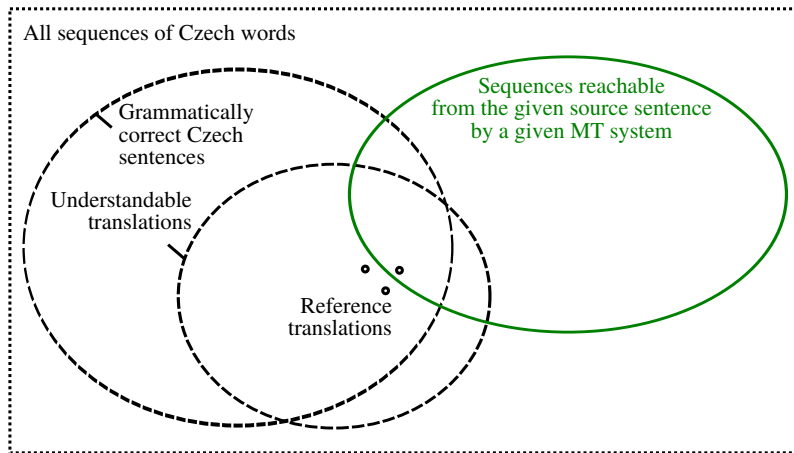
Space of Possible Translations

For a single English source sentence, we can consider:



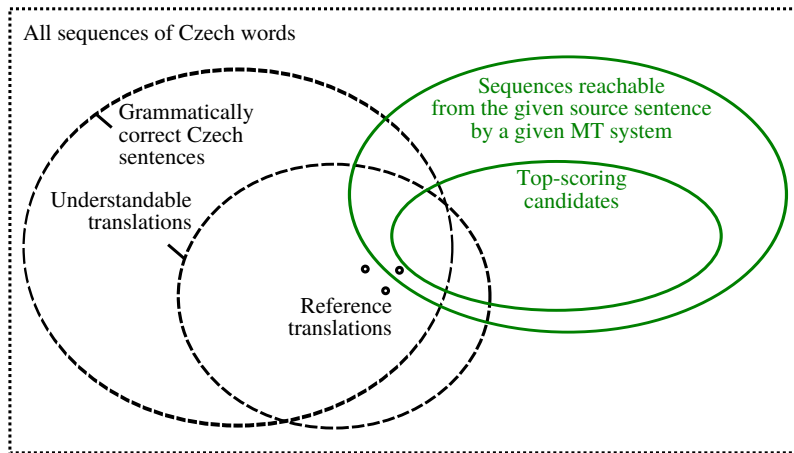
Space of Possible Translations

For a single English source sentence, we can consider:



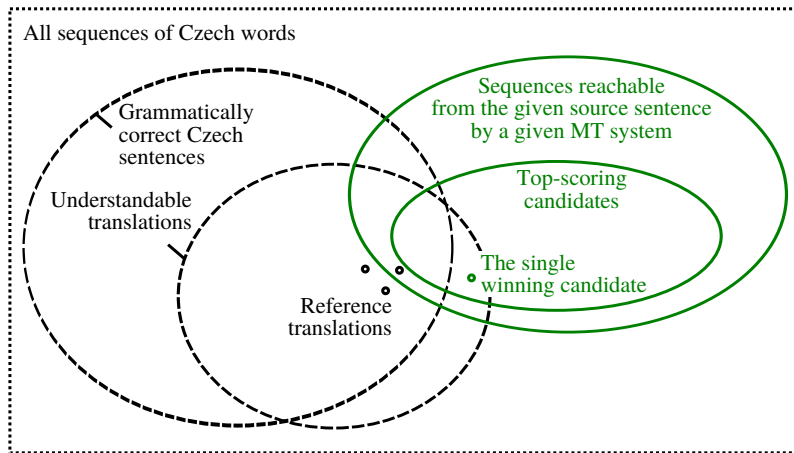
Space of Possible Translations

For a single English source sentence, we can consider:



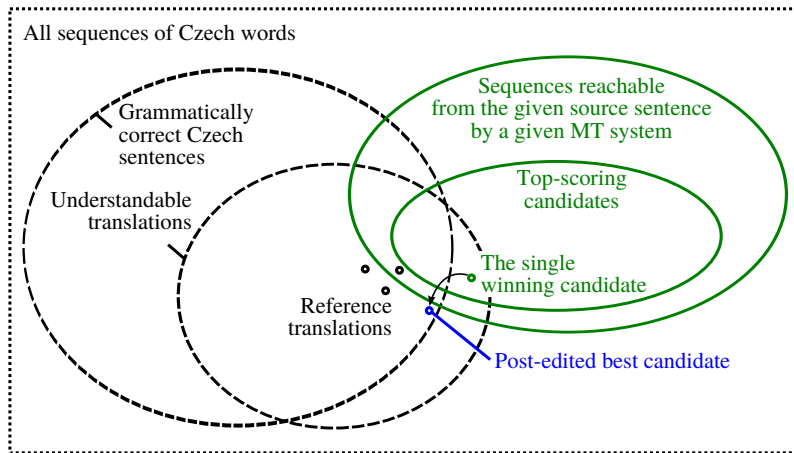
Space of Possible Translations

For a single English source sentence, we can consider:



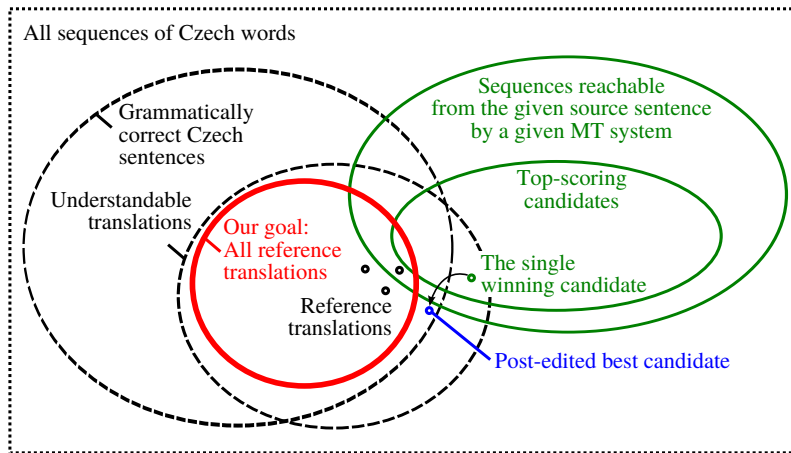
Space of Possible Translations

For a single English source sentence, we can consider:



Space of Possible Translations

For a single English source sentence, we can consider:



How Many Good Translations Are There?

How many good translations the following sentence has?

And even though he is a political veteran, the Councilor Karel Brezina responded similarly.

Data: <https://ufal.mff.cuni.cz/grants/deprefset> (Bojar et al., 2013)

How Many Good Translations Are There?

Examples of 71 thousand correct translations of the English:

And even though he is a political veteran, the Councilor Karel Brezina responded similarly.

A ačkoli ho lze považovat za politického veterána, radní Brezina reagoval obdobně.

Ač ho můžeme prohlásit za politického veterána, reakce radního Karla Breziny byla velmi obdobná.

A i přestože je politický matador, radní Karel Brezina odpověděl podobně.

A přestože je to politický veterán, velmi obdobná byla i reakce radního K. Breziny.

A radní K. Brezina odpověděl obdobně, jakkoli je politický veterán.

A třebaže ho můžeme považovat za politického veterána, reakce Karla Breziny byla velmi podobná.

Byť ho lze označit za politického veterána, Karel Brezina reagoval podobně.

Byť ho můžeme prohlásit za politického veterána, byla i odpověď K. Breziny velmi podobná.

K. Brezina, i když ho lze prohlásit za politického veterána, odpověděl velmi obdobně.

Odpověď Karla Breziny byla podobná, navzdory tomu, že je politickým veteránem.

Radní Brezina odpověděl velmi obdobně, navzdory tomu, že ho lze prohlásit za politického veterána.

Data: <https://ufal.mff.cuni.cz/grants/deprefset> (Bojar et al., 2013)

Scoring Techniques

Black-box: Judging hypotheses produced by MT systems:

- **Adequacy** and **fluency** of whole sentences.
Somewhat revisited under the name **Direct assessment (DA)**.
- **Relative ranking (RR) of full sentences** by several MT systems:
Longer sentences hard to rank. Candidates incomparably poor/good.
- **Ranking of constituents** (parts of sentences). Overall coherence not evaluated.
- **Comprehension test:** Blind editing+correctness check.
- **Task-based:** Does MT output help as much as the original?
Do I dress appropriately given a translated weather forecast?

Gray-box: Analyzing errors in systems' output.

- **HMEANT**, **HUME**: Is the core event structure preserved?
- **MQM**: Multi-dimensional quality metrics.
- **ESA**: Error Span Annotation.

Glass-box: System-dependent: Does this component work?

Direct Assessment: Adequacy

Graham et al. (2013) propose a simple continuous scale:

- To what extent MT adequately expresses the meaning of REF?

This HIT consists of 100 English assessments. You have completed 0.

Read the text below. How much do you agree with the following statement:

The black text adequately expresses the meaning of the gray text in English.

To snobs like me who declare that they'd rather play sports than watch them, it's hard to see the appeal of watching games rather than taking up a controller myself.

Snob like me, who say that it is better to be in sports than watching him, it is hard to understand the appeal of having to watch the game, rather than to take a joystick in hand.

0 %



100 %

- ⊕ After ~ 15 judgements, each annotator stabilizes.
- ⊖ Interpretable by averaging over many judgements of many people.
- ⊖ 30–70(!)% of participating Turkers unreliable.
- ⊖ Too few non-English speakers on Amazon Mechanical Turk.

Direct Assessment: Fluency

DA for fluency:

- To what extent the MT is fluent English?
- The source or reference are not shown at all.
- Fluency used only to break ties in adequacy.

This HIT consists of 100 English assessments. You have completed 18.

Read the text below. How much do you agree with the following statement:

The text is fluent English.

Mario Idol after what problems back in the squad sat out first.

0 %



100 %

NEXT

Caveats! MT Surpassing Humans: 2018

- WMT 2018 English-to-Czech news translation results: (Bojar et al., 2018)

	Ave. %	Ave. z	System
1	84.4	0.667	CUNI-Transformer
2	79.8	0.521	UEDIN
	78.6	0.483	Professional Translation
4	68.1	0.128	ONLINE-B
5	59.4	-0.178	ONLINE-A
6	54.1	-0.354	ONLINE-G

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Caveats:

- Humans translated whole documents, MT individual segments.
 - Evaluation was done for individual segments.

SRC-Based Doc-Level DA

Document

WMT19DocSrcDA #202:Document
#Independent.Z26349-10

English → German (deutsch)

Below are the sentences you have just rated as a single document. Please state how much you agree that:

The black text adequately expresses the meaning of the gray text in German (deutsch).

Russian Grand Prix: Lewis Hamilton closes in on world title after team orders hand him win over Sebastian Vettel It became clear from the moment that Valtteri Bottas qualified ahead of Lewis Hamilton on Saturday that Mercedes' team orders would play a large part in the race. From pole, Bottas got a good start and almost hung Hamilton out to dry as he defended his position in the first two turns and invited Vettel to attack his teammate. Vettel went into the pits first and left Hamilton to run into the traffic at the tail of the pack, something which should have been decisive. The Mercedes pitted a lap later and came out behind Vettel, but Hamilton went ahead after some wheel-to-wheel action that saw the Ferrari driver reluctantly leave the inside free at risk of holding out after a double-move to defend on the third corner. Max Verstappen started from the back row of the grid and was in seventh by the end of the first lap on his 21st birthday. He then led for a large part of the race as he held onto his tyres to target a quick finish and overtake Kimi Raikkonen for fourth. He eventually came into the pits on the 44th lap but was unable to increase his pace in the remaining eight laps as Raikkonen took fourth. It's a difficult day because Valtteri did a fantastic job all weekend and was a real gentleman told let me by. The team have done such an exceptional job to have a one two," said Hamilton.

— Source text

Großer Preis von Russland: Lewis Hamilton schließt auf Weltmeistertitel ein, nachdem ihm das Team den Sieg über Sebastian Vettel überlassen hat Es wurde von dem Moment an klar, dass Valtteri Bottas sich vor Lewis Hamilton am Samstag qualifiziert hatte, dass die Teamaufträge von Mercedes eine große Rolle im Rennen spielen würden. Von der Pole aus erwischte Bottas einen guten Start und ließ Hamilton fast trocken, als er seine Position in den ersten beiden Kurven verteidigte und Vettel einlud, seinen Teamkollegen anzugreifen. Vettel ging zuerst in die Gruben und verließ Hamilton, um am Rucksack in den Verkehr zu geraten, was entscheidend gewesen sein sollte. Der Mercedes drehte eine Runde später und kam hinter Vettel, aber Hamilton ging nach einigen Rad-an-Rad-Aktion, die sah, dass der Ferrari-Fahrer widerwillig verlassen die Innenseite frei in Gefahr zu halten, nach einem Doppelschlag auf der dritten Ecke zu verteidigen. Max Verstappen startete aus der hinteren Startreihe und wurde am Ende der ersten Runde an seinem 21. Geburtstag Siebter. Er führte dann für einen großen Teil des Rennens, als er auf seinen Reifen hielt, um ein schnelles Ziel zu erreichen und Kimi Räikkönen zum vierten Mal zu überholen. In der 44. Runde kam er schließlich in die Box, konnte aber sein Tempo in den verbleibenden acht Runden nicht erhöhen, da Räikkönen den vierten Platz belegte. Es ist ein schwieriger Tag, denn Valtteri hat das ganze Wochenende einen fantastischen Job gemacht und war ein echter Gentleman, der mir gesagt hat. Das Team hat so einen außergewöhnlichen Job gemacht, um ein, zwei zu haben", sagte Hamilton.

— Candidate translation

0% | | | 100%

Reset

Submit

- ⊖ Mental overload.
- ⊖ Too few scores collected ⇒ Difficult to get statistical significance.

SRC-Based Pseudo-Doc-Aware DA

- Score sentences using DA one by one.
- *In the original order* (i.e. not shuffled).

⇒ Mentally manageable.

Problems of the first run at WMT19 (Barrault et al., 2019):

- No way to go back to previous sentences.
- All sentences in a row must come from the same MT system.
- No longer independent probes (violating statistical assumptions).

Recent Results: MT Surpassing Humans: 2019

English→Czech

	Ave.	Ave. z	System
1	91.2	0.642	Professional Translators
2	86.0	0.402	CUNI-DOCTRANSFORMER-T2T
	86.9	0.401	CUNI-TRANSFORMER-T2T-2018
	85.4	0.388	CUNI-TRANSFORMER-T2T-2019
5	81.3	0.223	CUNI-DOCTRANSFORMER-MARIAN
	80.5	0.206	UEDIN
7	70.8	-0.156	ONLINE-Y
	71.4	-0.195	TARTUNLP-C
9	67.8	-0.300	ONLINE-G
10	68.0	-0.336	ONLINE-B
11	60.9	-0.594	ONLINE-A
12	59.3	-0.651	ONLINE-X

Recent Results: MT Surpassing Humans: 2019

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1	91.2	0.642	Professional Translators
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	86.9	0.401	CUNI-TRANSFORMER-T2T-2018
	85.4	0.388	CUNI-TRANSFORMER-T2T-2019
5	81.3	0.223	CUNI-DOCTRANSFORMER-MARIAN
	80.5	0.206	UEDIN
7	70.8	-0.156	ONLINE-Y
	71.4	-0.195	TARTUNLP-C
9	67.8	-0.300	ONLINE-G
10	68.0	-0.336	ONLINE-B
11	60.9	-0.594	ONLINE-A
12	59.3	-0.651	ONLINE-X

English→German

Ave.	Ave. z	System
90.3	0.347	Facebook-FAIR
93.0	0.311	Microsoft-WMT19-sent-doc
92.6	0.296	Microsoft-WMT19-doc-level
90.3	0.240	Professional Translation
87.6	0.214	MSRA-MADL
88.7	0.213	UCAM
89.6	0.208	NEU
87.5	0.189	MLLP-UPV
87.5	0.130	eTranslation
86.8	0.119	dfki-nmt
84.2	0.094	online-B
... 10 more systems here ...		
76.3	-0.400	online-X
43.3	-1.769	en-de-task

SRC-Based Doc-Aware 10-RankME

Mix of all:

- Two or more systems considered.
- Whole document shown.
- A section of 10 consecutive sentences scored in (1) adequacy, (2) fluency, (3) overall.

⇒ Combines relative, absolute, doc-level, sent-level.

⊖ Very time-consuming.

	G	H	I	J	K	L	M	N	O	P
1										
	Source	Translation1	T1_adequacy	T1_fluency	Translation2		T2_adequacy	T2_fluency	Optional comment	
160	"And we're protecting our shareholders from employment litigation."									
161	Companies started taking ethics, values and employee engagement more seriously in 2002 after accounting firm Arthur Andersen collapsed because of ethical violations from the Enron scandal, Quisen said.									
170	But it wasn't until "social media came into its own" that companies realized they couldn't stop their dirty laundry from going viral online.									
171	"Prior to using technology to monitor ethics, people used hope as a strategy," he said.									
172	Both Glint and Conscient offer their software as a service, charging companies recurring fees to use their products.									
173	It's a business model and opportunity that has the approval of venture capital investors, who have propped up both start-ups.	Je to obchodní model a příležitost, kterou schvalují odvětví kapitáloví investoři, jenž podpořili oba start-upy.			Je to obchodní model a příležitost, která má souhlas investorů rizikového kapitálu, kteří podpořili oba start-upy.				T1: chybný překlad termínu "venture capital"	
174	Conscient raised \$10 million in funding in February from firms such as Sapphire Ventures and Tola Capital, bringing its total capital raised to \$47 million.	Conscient vybrali v rámci své únorové kampaně od firem jako Sapphire Ventures a Tola Capital celkově 10 milionů \$ a nakonec si odnesli kapitál ve výši 47 milionů \$.	7	6	Conscient získal v únoru finanční prostředky ve výši 10 milionů dolarů od firem jako Sapphire Ventures a Tola Capital, čímž se jeho celkový kapitál zvýšil na 47 milionů dolarů.		10	10		
175	Glint secured \$10 million in November from Bessemer Venture Partners, bringing its total funding to \$60 million.	Glint získal v listopadu 10 milionů \$ od Bessemer Venture Partners a v průběhu celé kampaně získal 60 milionů \$.	3	4	Glint získal v listopadu 10 milionů dolarů od společnosti Bessemer Venture Partners, čímž jeho celkové financování dosáhlo 60 milionů dolarů.		10	10		
176	These investments hardly come as a surprise, given the interconnected nature of companies, culture and venture capital.	Tyto investice jsou stábi překvapující vzhledem k vzájemné povaze společností, kultury a rizikového kapitálu.	5	4	Tyto investice nejsou vzhledem k propojenosti společností, kultury a rizikového kapitálu žádným překvapením.		10	10		
177	There's a growing body of research showing today's employees expect more from their workplaces than before.	Narůstající počet výzkumů jasně potvrzuje, že dnešní zaměstnanci očekávají od svého pracoviště více než dříve.	3	4	Roste množství výzkumů, které ukazují, že dnešní zaměstnanci očekávají od svých pracovišť více než dříve.		10	10		
178	In competitive markets such as Silicon Valley, high salaries and interesting projects are merely table stakes.	A na konkurenčních trzích, jakým je např. Silicon Valley, nejsou hlavní výhodou vysoké platy a zajímavé projekty.	5	5	Na konkurenčních trzích, jako je Silicon Valley, jsou vysoké platy a zajímavé projekty pouhými stávkami u stolu.		7	8	7	problém: význam termínu "table stakes"
179	Employees want to feel that they're accepted and valued and that they're giving their time to a company with a positive mission.	Zaměstnanci chtějí vnímat, že jsou přijímáni a oceňováni a že věnují svůj čas společnosti, která usiluje o pozitivní posláním.	6	5	Zaměstnanci chtějí mít pocit, že jsou přijímáni a oceňováni a že věnují svůj čas společnosti s pozitivním posláním.		10	10	5	

Relative Ranking of Sentences

Defying the shadows, Anto descends the crater and lights the path with a small torch attached to the helmet he bought with his money.

I přes okolní tmu fará Anto do kráteru a osvětluje si cestu malou svítilnou, kterou má připevněnou na helmě a sám si ji za své peníze koupil.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Vzdoruje stínům Anto, sestupuje z kráteru a svítí cestu s malou pochodní připojenou k helmě, kterou koupil ze svých peněz.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Vzpírá se stínům, Anto sestupuje kráter a osvětí cestu malou baterkou spojenou s helmou, kterou on koupil s jeho penězi.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Odolává stíny, Anto snáší kráter a osvětlí cestu s malou pochodně na helmu, koupil za své peníze.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Navzdory stínům anto, sestupuje z kráteru a svítí na cestu s malou pochodně připevněnou na helmu, kterou si koupil ze svých peněz.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Popírání stínovým zpravodajům, Anto nezávládne se crater a svítlidla cestu s malou pochodně oddání helmu koupil s jeho peníze.

Submit

Relative Ranking (Eye-Trackered)

Defying the shadows, Anto descends the crater and lights the path with a small torch attached to the helmet he bought with his money.

pres okolní tmou tárá Anto do kráteru a osvětluje si cestu malou svítilnou, kterou má připevněnou na helmě a sám si ji za své peníze koupil.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Vzpírá se stínům, Anto sestupuje z kráteru a svítí cestu s malou pochodní připojenou k helmě, kterou koupil ze svých peněz.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Vzpírá se stínům, Anto sestupuje kráter a osvětí cestu malou baterkou spojenou s helmou, kterou on koupil s jeho penězi.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Odolává stíny, Anto snáší kráter a osvětí cestu s malou pochodní na helmě, koupil za své peníze.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Navzdory stínům anto, sestupuje z kráteru a svítí na cestu s malou pochodní připevněnou na helmě, kterou si koupil ze svých peněz.

Best ← Rank 1 Rank 2 Rank 3 Rank 4 Rank 5 → Worst

Popírá stínovým zpravodajům, Anto nezávadně se crater a svítilna cestu s malou pochodní oddání helmě koupil s jeho peníze.

Submit

Project suggestion: Analyze the recorded data: path patterns / errors in words.

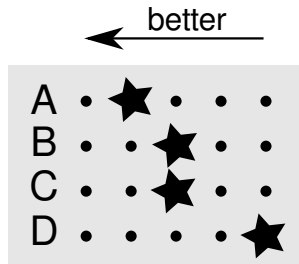
Relative Ranking of Constituents

Source: Können die USA **ihre Besetzung aufrechterhalten**, wenn sie dem irakischen Volk nicht Nahrung, Gesundheitsfürsorge und andere grundlegende Dienstleistungen anbieten können?

Reference: Can the US sustain its occupation if it cannot provide food, health care, and other basic services to Iraq's people?

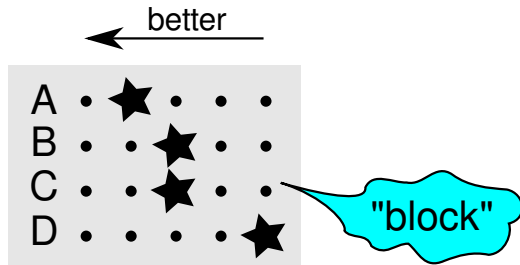
Translation	Rank
The United States can maintain its employment when it the Iraqi people not food, health care and other basic services on offer?.	☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 WorstBest
The US can maintain its occupation , if they cannot offer the Iraqi people food, health care and other basic services?	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☒ 5 WorstBest
Can the US their occupation sustained if it to the Iraqi people not food, health care and other basic services can offer?	☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5 WorstBest
Can the United States maintain their occupation , if the Iraqi people do not food, health care and other basic services can offer?	☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 WorstBest
The United States is maintained , if the Iraqi people, not food, health care and other basic services can offer?	☐ 1 ☒ 2 ☐ 3 ☐ 4 ☐ 5 WorstBest
Annotator: ccb Task: WMT07 German-English News Corpus	
Instructions: Rank each constituent translation from Best to Worst relative to the other choices (ties are allowed). Grade only the highlighted part of each translation. <i>Please note that segments are selected automatically, and they should be taken as an approximate guide. They might include extra words on either end that are not in the actual alignment, or miss words.</i>	

Interpreting Manual Ranks



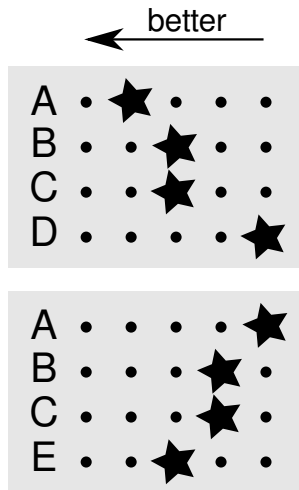
See also Bojar et al. (2011).

Interpreting Manual Ranks



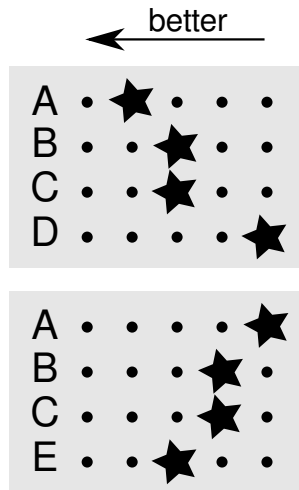
See also Bojar et al. (2011).

Interpreting Manual Ranks



See also Bojar et al. (2011).

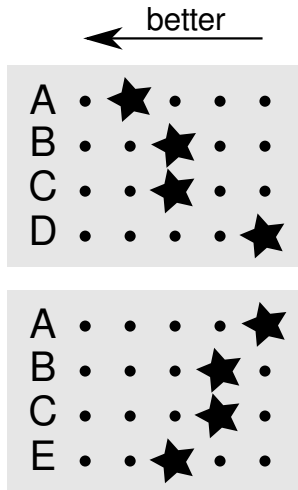
Interpreting Manual Ranks



Who Wins WMT?

See also Bojar et al. (2011).

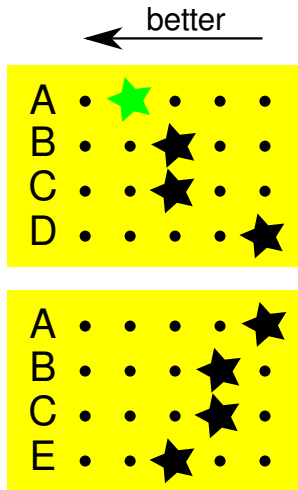
Interpreting Manual Ranks



[Systems] are ranked based on how frequently they were judged to be better than or equal to any other system.

See also Bojar et al. (2011).

Interpreting Manual Ranks

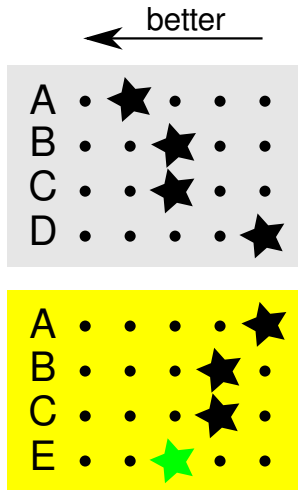


"≥ All in Block"

A: 1/2
B: 0/2
C: 0/2
D: 0/1
E: 1/1

See also Bojar et al. (2011).

Interpreting Manual Ranks

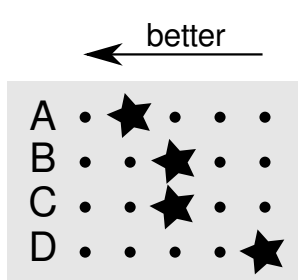


"≥ All in Block"

A: 1/2
B: 0/2
C: 0/2
D: 0/1
E: 1/1

See also Bojar et al. (2011).

Interpreting Manual Ranks



Simulated
Pairwise

"≥ All in Block"

A: 1/2

B: 0/2

C: 0/2

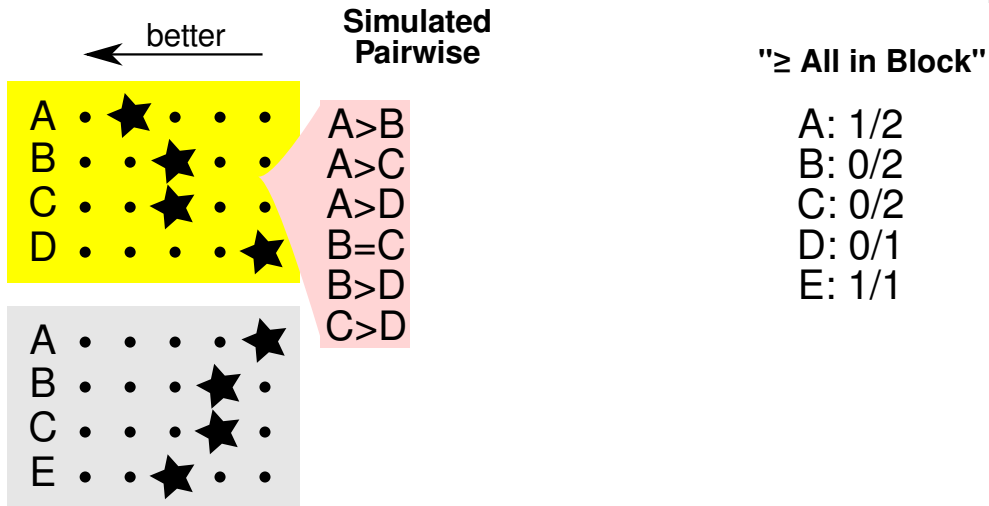
D: 0/1

E: 1/1



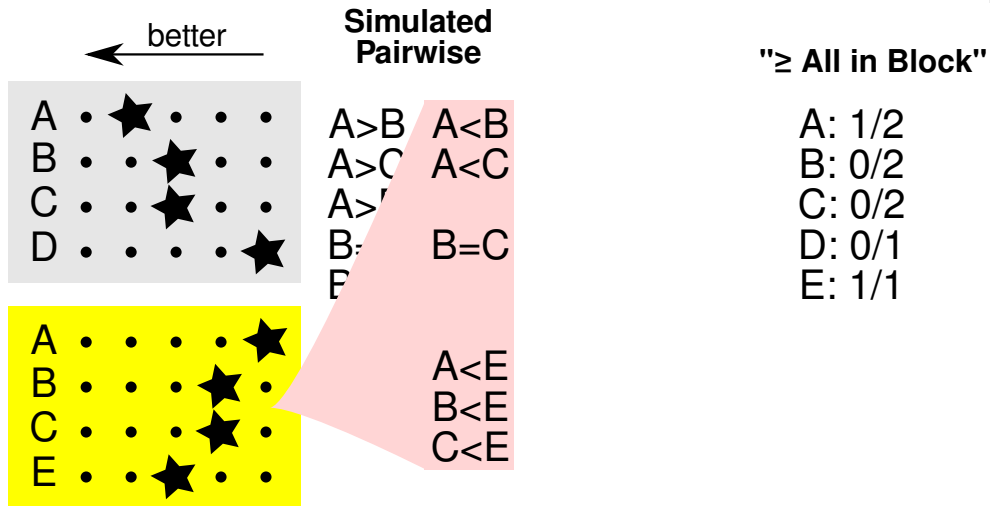
See also Bojar et al. (2011).

Interpreting Manual Ranks



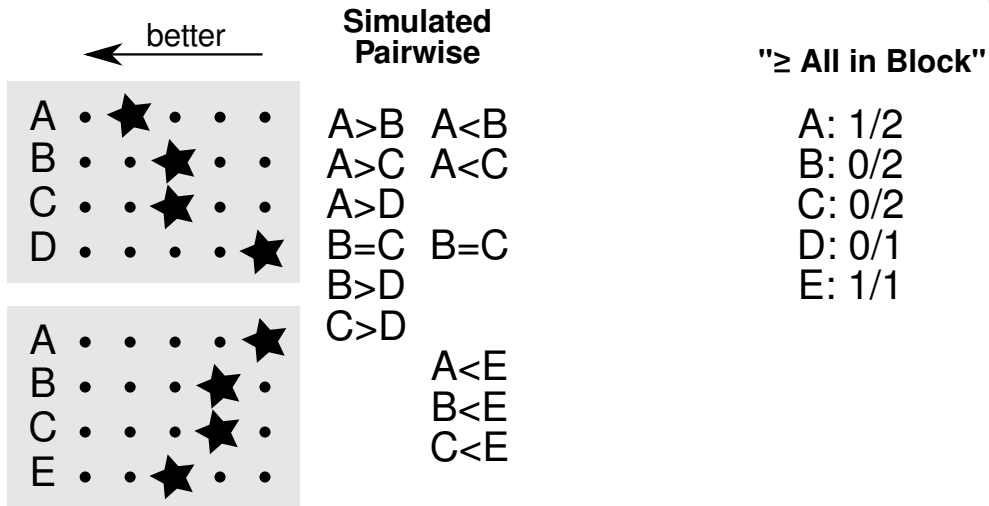
See also Bojar et al. (2011).

Interpreting Manual Ranks



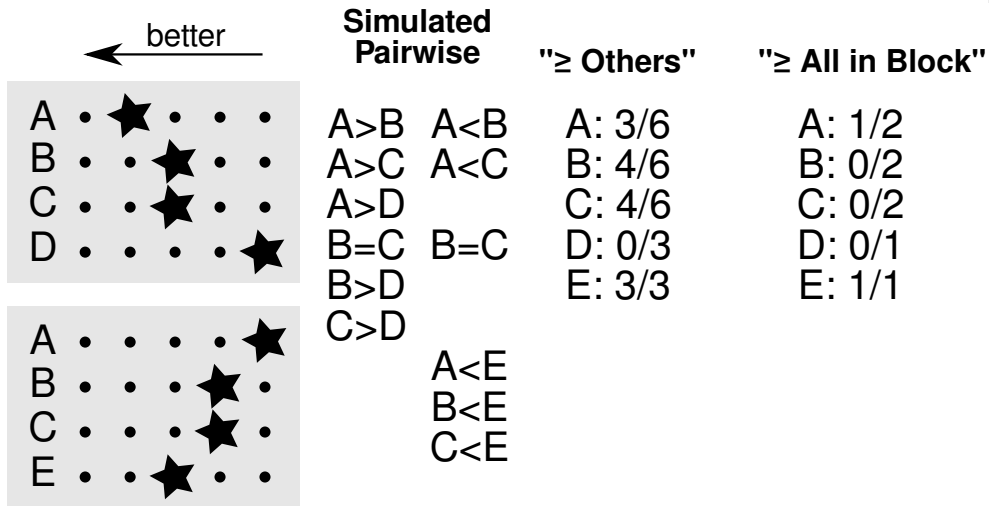
See also Bojar et al. (2011).

Interpreting Manual Ranks



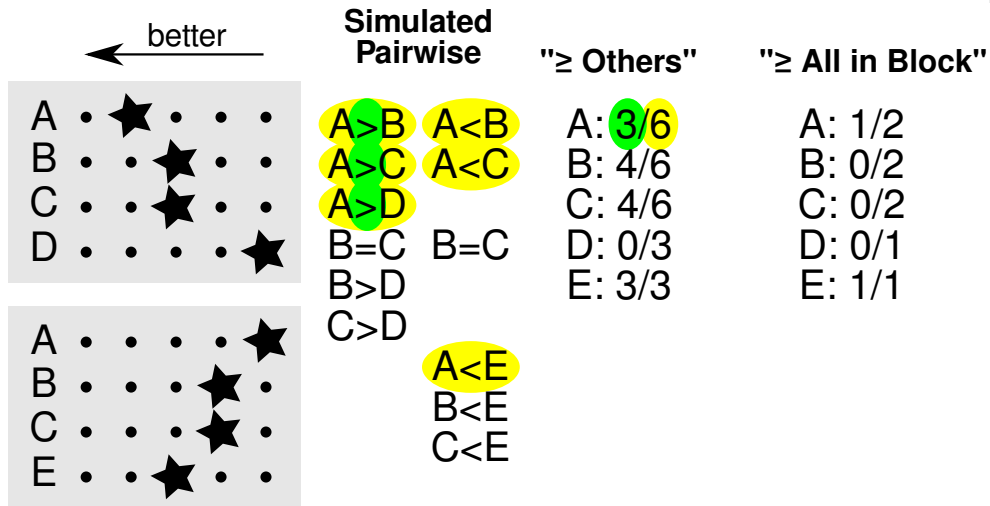
See also Bojar et al. (2011).

Interpreting Manual Ranks



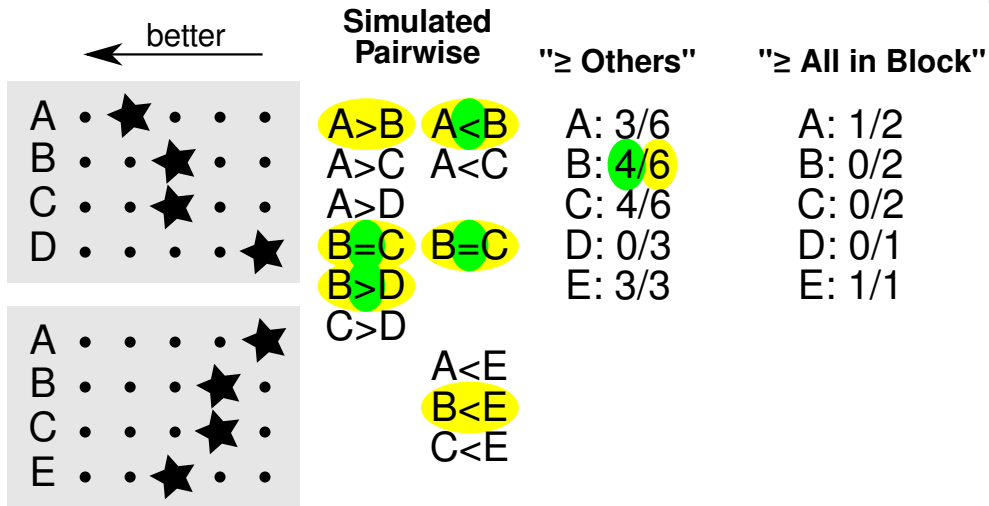
See also Bojar et al. (2011).

Interpreting Manual Ranks



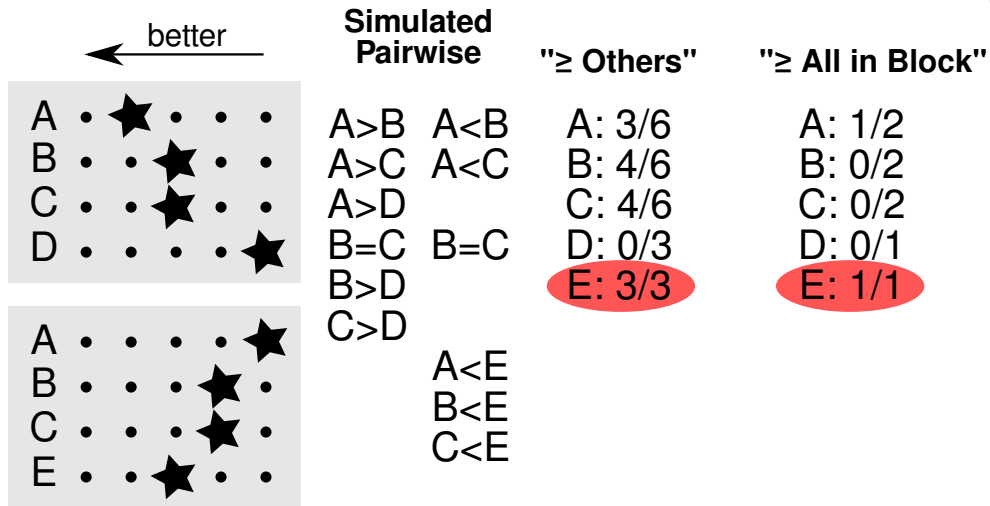
See also Bojar et al. (2011).

Interpreting Manual Ranks



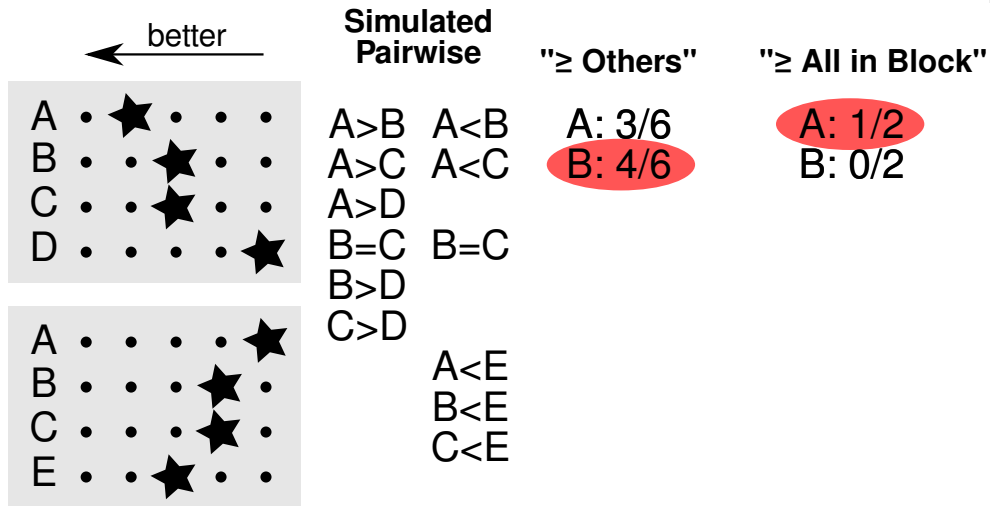
See also Bojar et al. (2011).

Interpreting Manual Ranks



See also Bojar et al. (2011).

Interpreting Manual Ranks



See also Bojar et al. (2011).

Comprehension 1/2 (Blind Editing)

Original: They are often linked to other alterations sleep as nightmares, night terrors, the nocturnal enuresis (pee in bed) or the sleepwalking, but it is not always the case.

Edit:

They are often linked to other sleep disorders, such as nightmares, night terrors, the nocturnal enuresis (bedwetting) or sleepwalking, but this is not always the case.

[Reset Edit](#)

- ☒ Edited.
☐ No corrections needed.
☐ Unable to correct.

Annotator: ccb **Task:** WMT09 Multisource-English News Editing

Instructions:

Correct the translation displayed, making it as fluent as possible. If no corrections are needed, select "No corrections needed." If you cannot understand the sentence well enough to correct it, select "Unable to correct."

Comprehension 2/2 (Judging)

Source: Au même moment, les gouvernements belges, hollandais et luxembourgeois ont en parti nationalisé le conglomérat européen financier, Fortis. Les analystes de Barclays Capital ont déclaré que les négociations frénétiques de ce week end, conclues avec l'accord de sauvetage" semblent ne pas avoir réussi à faire revivre le marché".

Alors que la situation économique se détériorasse, la demande en matières premières, pétrole inclus, devrait se ralentir.

"la prospective d'équité globale, de taux d'intérêt et d'échange des marchés, est devenue incertaine" ont écrit les analystes de Deutsche Bank dans une lettre à leurs investisseurs."

"nous pensons que les matières premières ne pourront échapper à cette contagion.

Reference: Meanwhile, the Belgian, Dutch and Luxembourg governments partially nationalized the European financial conglomerate Fortis.

Analysts at Barclays Capital said the frantic weekend negotiations that led to the bailout agreement "appear to have failed to revive market sentiment."

As the economic situation deteriorates, the demand for commodities, including oil, is expected to slow down.

"The outlook for global equity, interest rate and exchange rate markets has become increasingly uncertain," analysts at Deutsche Bank wrote in a note to investors.

"We believe commodities will be unable to escape the contagion.

Translation	Verdict
While the economic situation is deteriorating, demand for commodities, including oil, should decrease.	☒ Yes ☐ No
While the economic situation is deteriorating, the demand for raw materials, including oil, should slow down.	☒ Yes ☐ No
Alors que the economic situation deteriorated, the request in rawmaterial enclosed, oil, would have to slow down.	☐ Yes ☒ No
While the financial situation damaged itself, the first matters affected, oil included, should slow down themselves.	☐ Yes ☒ No
While the economic situation is depressed, demand for raw materials, including oil, will be slow.	☐ Yes ☒ No
Annotator: ccb Task: WMT09 French-English News Edit Acceptance	
Instructions: Indicate whether the edited translations represent fully fluent and meaning-equivalent alternatives to the reference sentence. The reference is shown with context, the actual sentence is bold .	

Quiz-Based Evaluation (1/1)

An approximation of task-based evaluation.

Preparation: English texts and Czech yes/no questions:

- We *found* English text snippets hopefully by native speakers.
- We equipped each snippet with 3 yes/no questions in Czech.

3 different snippet lengths (1..3 sents.), 4 different topics:

- Meeting: when, where, how often, with whom, ...
- Directions: driving/walking instructions, finding buildings, ...
- Basic quizzes: maths, physics, biology, ... simple questions.
- Politics/News: elections chances, affairs, finance news, ...

Annotation: Given machine-translated snippet, answer the questions.

Quiz-Based Evaluation (2/2)

Moses 2007

Na provoz světla na roundabout, obrátit levice a projet ballymun. Otočit vlevo na křižovatce. ballymun / Collins Avenue Road Dcu je umístěna na Collins 500m na pravém boku Avenue.

Google 16.2.2010

Na semaforech na kruhový objezd, odbočit doleva a jet přes Ballymun. Odbočit vlevo na Collins Avenue / Ballymun silniční křižovatky. DCU se nachází na Collins Avenue 500 m na pravé straně.

Zaškrtněte pravdivá tvrzení:

1. DCU leží na Collins Avenue.
2. V daném městě mají na kruhových objezdech zřejmě semaforey.
3. Při příjezdu budete mít DCU po levé straně.

Original: At the traffic lights on the roundabout, turn left and drive through Ballymun. Turn left at the Collins Avenue/Ballymun Road crossroads. DCU is located on Collins Avenue 500m on the right hand side.

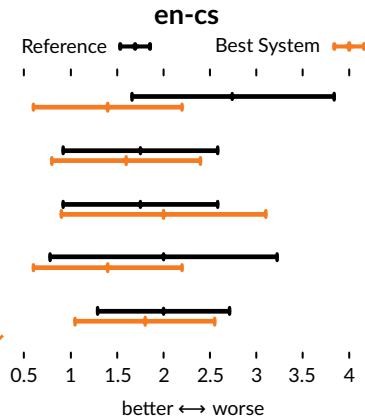
Correct answer: yyn

Maturita (GCSE)-Like (Vojtěchová et al., 2019)

Manual evaluation by domain experts, scoring in categories:

1. Language Resources – Spelling and Morphology
2. Vocabulary – Adequacy of Terms Used
3. Vocabulary – Clarity of the Text in Terms of Used Words
4. Syntax and Word Order
5. Coherence and Overall Understanding of the Text

plotted as average rank for better comparability



Superhuman MT Translating Agreements?

Supplement No. 1 to the agreement on the sublease the apartment, of 13th May 2016
On the day, month and year written below Marta Burešová, pers. no. 695604/3017
Address: Radimova 8, Prague 6, 169 00 as the tenant on the one hand (Hereinafter referred to as "the tenant") and Karolína Černá, pers. no. 136205/891 Address: Alfrédova 13, Praha 4, 142 00 As a lessee on the other (Hereinafter referred to as "the lessee") collectively also referred to as "the Contracting parties" have agreed on this Supplement No. 1 to the Agreement on the sublease the apartment, of 13th May 2016 (hereinafter referred to as the "Supplement No. 1")

I. Introductory Provisions

On 13th May 2016, the tenant and the lessee closed the Agreement on the sublease of the apartment, under which the tenant let the lessee use the apartment No. 4 (area 49 m²) of size 1+1/L in the ground floor of the house in Prague 4, Alfrédova 13, ...

(Vojtěchová et al., 2019)

Superhuman MT Translating Agreements?

Dodatek č. 1 ke smlouvě o podnájmu bytu ze dne 13. května 2016

V den, měsíc a rok níže napsané Marta Burešová, pers. no.

695604/3017 Adresa: Radimova 8, Praha 6, 169 00 jako nájemce na
jedné straně (dále jen „nájemce“) a Karolína Černá, pers. no.

136205/891 Adresa: Alfrédova 13, Praha 4, 142 00 jako nájemce na
straně druhé (dále jen „nájemce“) společně označované také jako
„smluvní strany“ se dohodly na tomto dodatku č. 1 ke smlouvě o
podnájmu, dále jen „nájemní smlouva“, dále jen „13. května 2016“).

I. Úvodní ustanovení

Dne 13. května 2016 nájemce a nájemce uzavřeli smlouvu o dalším
pronájmu bytu, podle níž nájemce pronajímá nájemci byt č. 4 (plocha
49 m²) o velikosti 1+1/I v přízemí domu v Praze 4, Alfrédova 13, ...

Superhuman MT Translating Agreements?

Dodatek č. 1 ke smlouvě o podnájmu bytu ze dne 13. května 2016

V den, měsíc a rok níže napsané Marta Burešová, pers. no.

695604/3017 Adresa: Radimova 8, Praha 6, 169 00 jako **nájemce na jedné straně** (dále jen „**nájemce**“) a Karolína Černá, pers. no.

136205/891 Adresa: Alfrédova 13, Praha 4, 142 00 jako **nájemce na straně druhé** (dále jen „**nájemce**“) společně označované také jako „smluvní strany“ se dohodly na tomto dodatku č. 1 ke smlouvě o podnájmu, dále jen „**nájemní smlouva**“, dále jen „13. května 2016“).

I. Úvodní ustanovení

Dne 13. května 2016 **nájemce a nájemce** uzavřeli smlouvu o dalším pronájmu bytu, podle níž **nájemce pronajímá nájemci** byt č. 4 (plocha 49 m²) o velikosti 1+1/I v přízemí domu v Praze 4, Alfrédova 13

HMEANT (Lo and Wu, 2011)

- Improved evaluation of adequacy compared to BLEU.
- Reduced human labour of HTER (Snover et al., 2006).

Essence: *Is the basic event structure understandable?*

(*Who did what to whom, when, where and why.*)

1. Identify semantic frames and roles in ref & hyp.
 - Manual (5–15 min of training) or automatic (shallow SRL).
2. Mark match/partial/mismatch of each predicate and each argument.
 - Manual.
3. Calculate prec & rec across all frames in the sentence.
4. Report f-score.

HMEANT Illustration: Motivation

It is hard to rank A vs. B (even if we know R is the ref.)



Finally, he stood in the center
of the referee Wolfgang Stark.



At the end of the day, was at the centre
of a referee Wolfgang Stark.



The referee Wolfgang Stark then garnered
some attention.

HMEANT Illustration: SRL

It is easier to mark roles of a single hypothesis.



Finally, he stood in the center
of the referee Wolfgang Stark.

HMEANT Illustration: SRL

It is easier to mark roles of a single hypothesis.



Finally, he **stood** in the center
Action of the referee Wolfgang Stark.

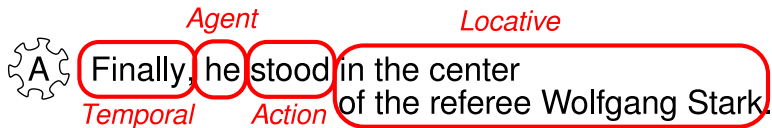
HMEANT Illustration: SRL

It is easier to mark roles of a single hypothesis.

 Finally, *Agent* he stood *Action* in the center
of the referee Wolfgang Stark.

HMEANT Illustration: SRL

It is easier to mark roles of a single hypothesis.



HMEANT Illustration: SRL

The same SRL is performed on the reference.



The referee Wolfgang Stark then garnered
some attention.

HMEANT Illustration: SRL

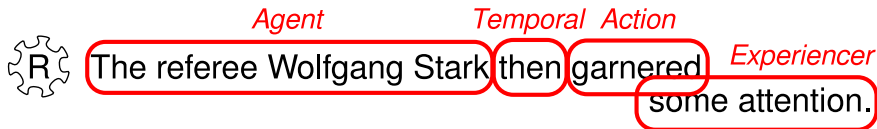
The same SRL is performed on the reference.



The referee Wolfgang Stark then *Action* garnered some attention.

HMEANT Illustration: SRL

The same SRL is performed on the reference.



HMEANT Illustration: Alignment

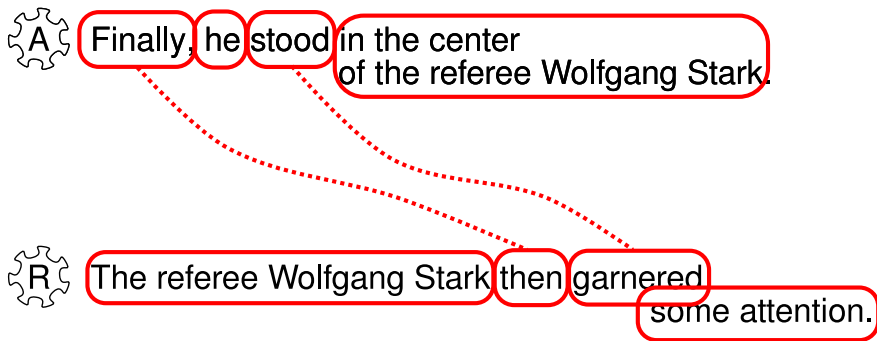
And finally, frames and role fillers are aligned.

 Finally, he stood in the center
of the referee Wolfgang Stark.

 The referee Wolfgang Stark then garnered
some attention.

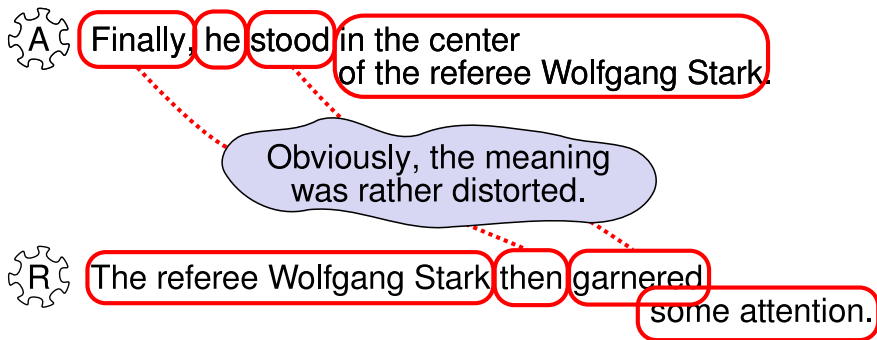
HMEANT Illustration: Alignment

And finally, frames and role fillers are aligned.



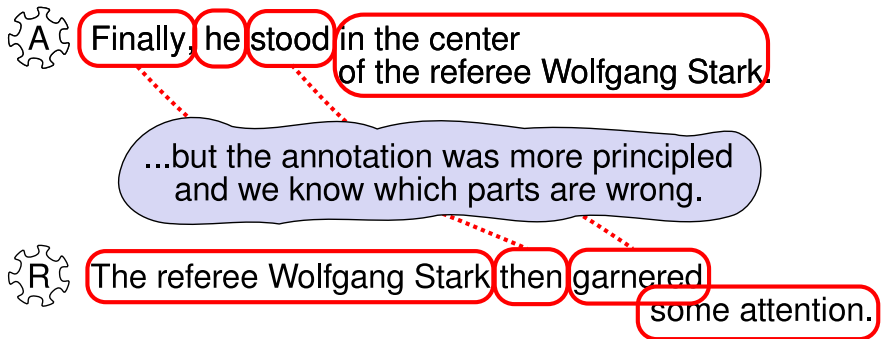
HMEANT Illustration

And finally, frames and role fillers are aligned.



HMEANT Illustration

And finally, frames and role fillers are aligned.



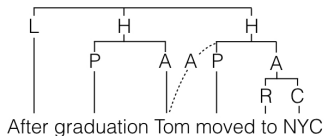
HUME

HUME (Birch et al., 2016) improves over HMEANT by:

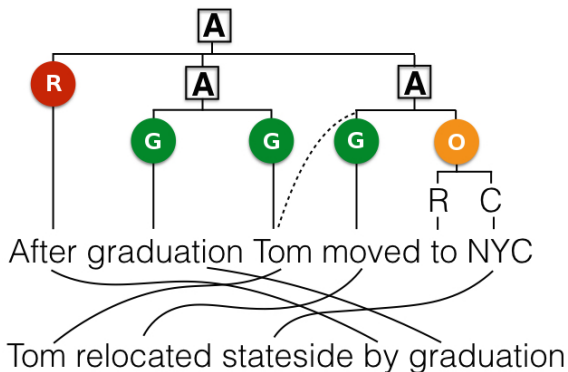
- using semantic trees (UCCA, Abend and Rappoport (2013)),
- using *source rather than reference*,
- using trees on the *source only*, not malformed hypothesis.

Two manual stages again:

1. Create UCCA tree for the source (can reuse for more systems!).
2. Label UCCA tree indicating how much was preserved by MT.

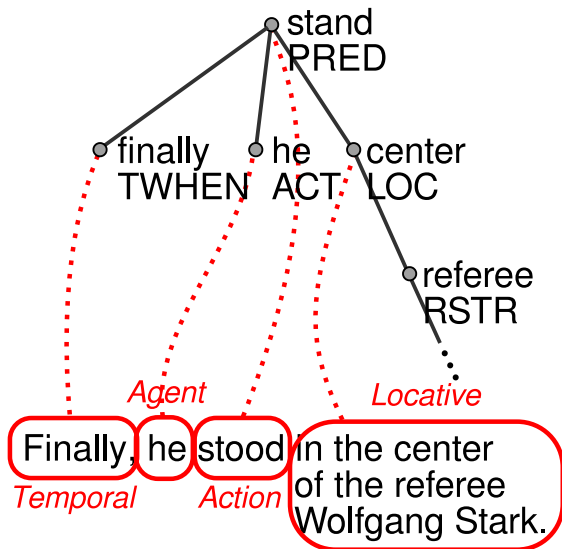


HUME Annotation



- Leafs get R/O/G (traffic lights): bad, mixed, good.
- Structure gets A/B: adequate, bad.

HMEANT/HUME are Close to FGD



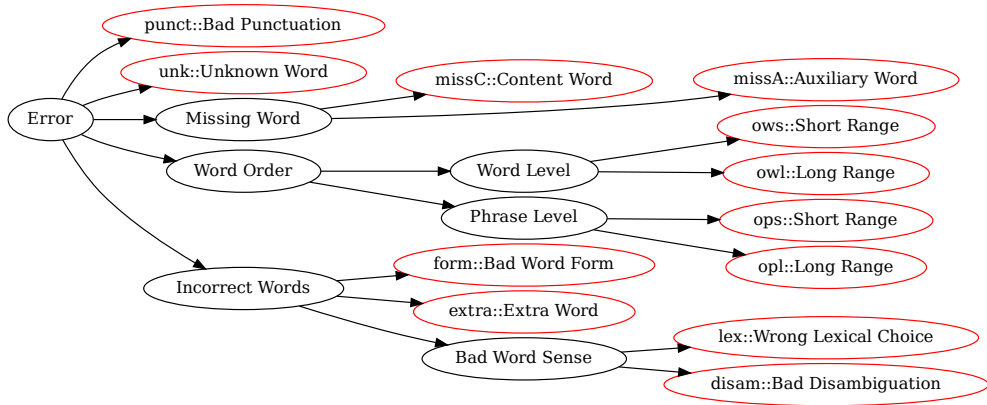
Project suggestion:

Use t-layer tools to:

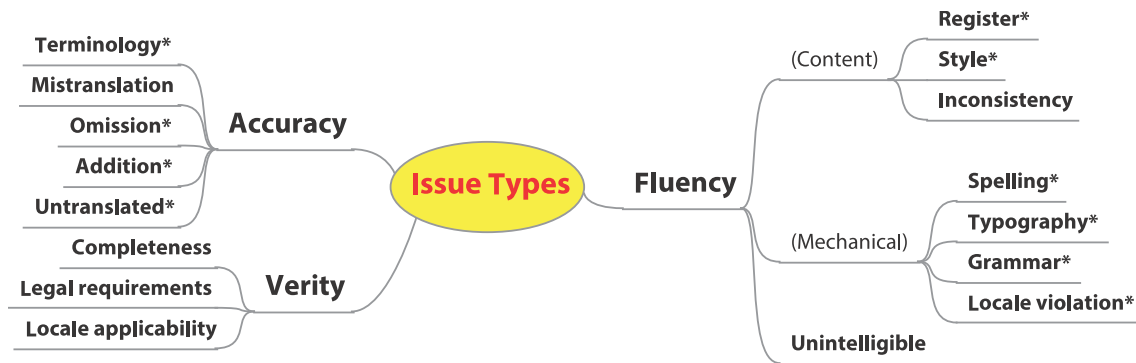
- Improve UCCA parser, or
- Automate: parse to UCCA or t-trees, predict R/O/G, A/B.

Evaluation by Flagging Errors

Classification of MT errors, following Vilar et al. (2006).

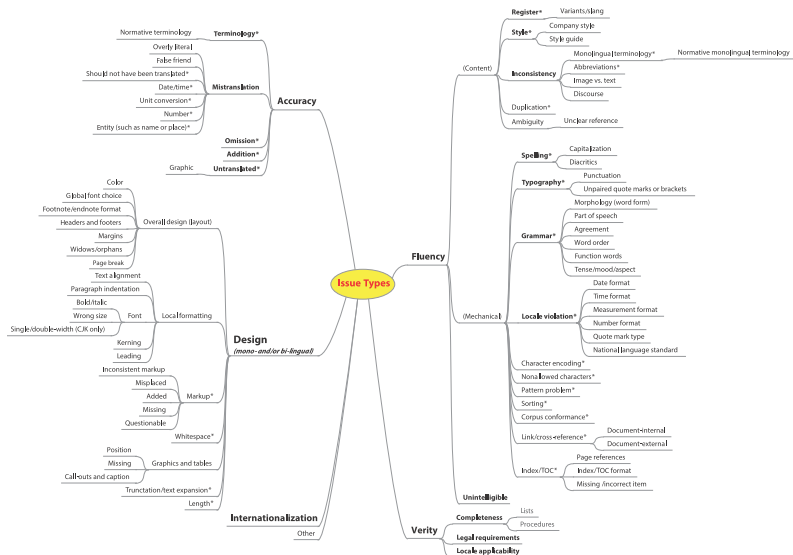


Standard MQM (Core)



(Lommel et al., 2014)

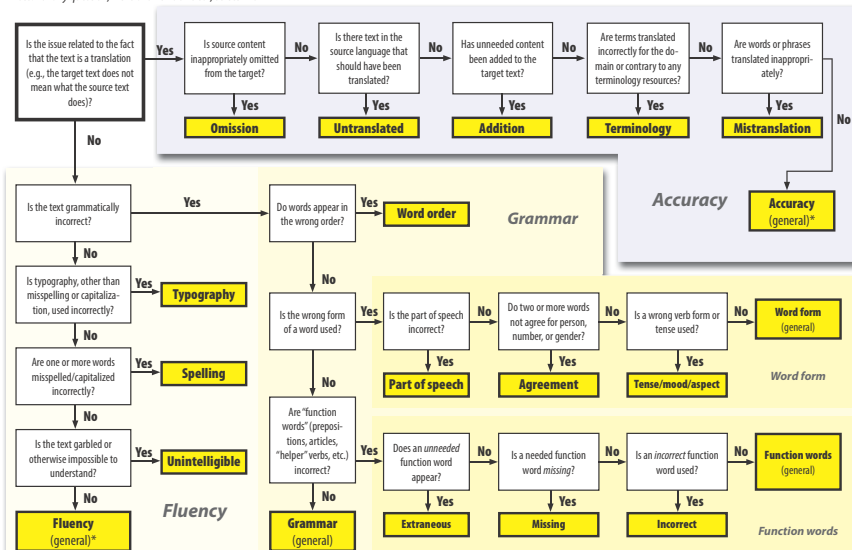
Standard MQM (Overkill)



MQM Decision Tree (Simplified)

MQM ANNOTATION DECISION TREE

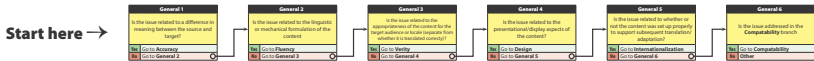
Note: For any question, if the answer is unclear, select "No"



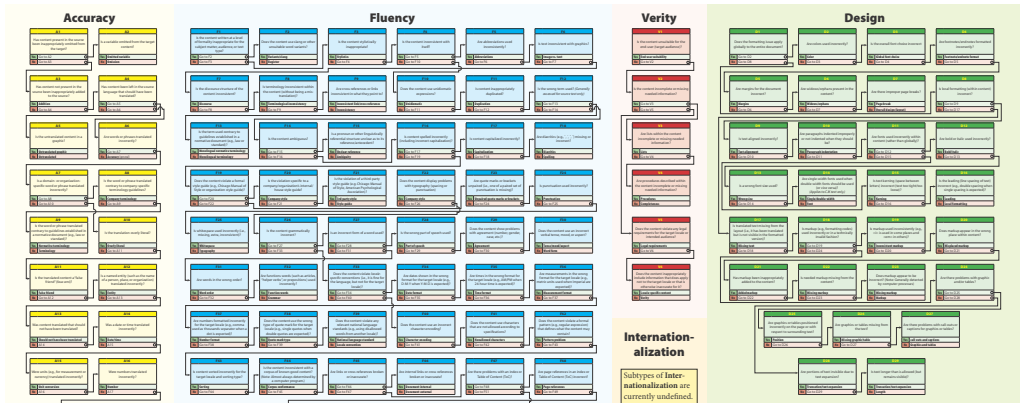
MQM Decision Tree (Full)

The Multidimensional Quality Metrics (MQM) Framework provides a hierarchical categorization of error types that occur in translated or localized products. Based on a detailed analysis of existing translation quality metrics, it provides a flexible typology of *issue types* that can be applied to analytic or holistic translation quality evaluation tasks. Although the full MQM issue tree (which, as of November 2014, contains 115 issue types categorized into five major branches) is not intended to be used in its entirety for any particular evaluation task, this overview chart presents a 'decision tree' suitable for selecting an issue type from it. In practical terms, however, an individual metric would have a smaller decision tree that covers just the issues contained in that metric.

To use the decision tree start with the first question and follow the appropriate answers until a specific issue type is reached.



Note: If the issue is found in the Computability branch it is probably not a translation product issue, but instead a process issue and would normally not be addressed in MJQI. If the issue is **Other**, it may not be a translation-related issue since translation-related issues would normally fall into one of the major branches.



Internationalization

Subtypes of Internationalization are currently undefined.

Error Span Annotation (ESA) at WMT25

- No errors types.
- Only minor/major.
- Overall DA score at the end.

Czech (čeština)

Plánujete svou cestu do Cambridge
Hledáte místo k pobytu. Hotel by měl být na východě a měl by být v levném cenovém rozmezí.
Hotel by měl zahrnovat bezplatné parkování.
Jakmile najdete hotel, který chcete, rezervujete ho pro 8 osob a 4 noci, počínaje čtvrtkem.
Ujistěte se, že dostanete referenční číslo.
Hledáte také místo k večeři. Restaurace by měla být ve stejném cenovém rozmezí jako hotel a měla by podávat indické jídlo.
Jakmile najdete restauraci, kterou chcete, rezervujete stůl pro stejnou skupinu lidí na 12:45 ve stejný den.
Ujistěte se, že dostanete referenční číslo.
Také chcete rezervovat taxi, které vás přepraví mezi těmito dvěma místy.
Chcete, aby dorazilo do restaurace včas.
Ujistěte se, že dostanete kontaktní číslo a typ vozu.

German (Deutsch)

Plánieren Sie Ihre Reise nach Cambridge
Sie suchen eine Unterkunft. Das Hotel sollte im Osten liegen und preiswert sein.
Das Hotel sollte kostenloses Parken beinhalten.
Sobald Sie das gewünschte Hotel gefunden haben, buchen Sie es für 8 Personen und 4 Nächte, beginnend am Donnerstag.
Stellen Sie sicher, dass Sie die Referenznummer erhalten.
Sie suchen auch einen Ort zum Abendessen. Das Restaurant sollte im gleichen Preisniveau wie das Hotel liegen und indisches Essen servieren.
Sobald Sie das gewünschte Restaurant gefunden haben, buchen Sie einen Tisch für die gleiche Gruppe von Personen um 12:45 Uhr am selben Tag.
Stellen Sie sicher, dass Sie die Referenznummer erhalten.
Sie möchten auch ein Taxi buchen, das Sie zwischen diesen beiden Orten befördert.
Sie möchten, dass es pünktlich zum Restaurant kommt.
Stellen Sie sicher, dass Sie die Kontaktnummer und den Fahrzeugtyp erhalten. **[MISSING]**

0%: Broken/poor 33%: Flawed 66%: Good 100%: Perfect

Error Flagging Example (already 2009)

Annotation rules:

- Mark/suggest as little as necessary.
- Compare to *source*, not to *reference*. Literal translation ok.
- Preserve white space. Don't add or remove word/line breaks.
- Only insert error labels followed by ::.
- For missing words, use _ instead of space, if necessary.

Src	Perhaps there are better times ahead.
Ref	Možná se tedy blýská na lepší časy.
	Možná, že extra:: tam jsou lepší disam:: krát lex:: dopředu.
	Možná extra:: tam jsou příhodnější časy vpředu.
missC::v_budoucnu	Možná form:: je lepší časy.
	Možná jsou lepší časy lex:: vpřed.

Results on WMT09 Dataset

	google	cu-bojar	pctrans	cu-tectomt	Total
Automatic: BLEU	13.59	14.24	9.42	7.29	—
Manual: Rank	0.66	0.61	0.67	0.48	—
disam	406	379	569	659	2013
lex	211	208	231	340	990
Total bad word sense	617	587	800	999	3003
missA	84	111	96	138	429
missC	72	199	42	108	421
Total missed words	156	310	138	246	850
form	783	735	762	713	2993
extra	381	313	353	394	1441
unk	51	53	56	97	257
Total serious errors	1988	1998	2109	2449	8544
ows	117	100	157	155	529
punct	115	117	150	192	574
...	...	...	...	...	...
tokenization	7	12	10	6	35
Total errors	2319	2354	2536	2895	10104

Contradictions in Manual Evaluation

Results for WMT10:

Evaluation Method	Google	CU-Bojar	PC Translator	TectoMT
$\geq$ others (WMT10 official)	70.4	65.6	62.1	60.1
$>$ others	49.1	45.0	49.4	44.1
Edits deemed acceptable [%]	55	40	43	34
Quiz-based evaluation [%]	80.3	75.9	80.0	81.5
Automatic: BLEU	0.16	0.15	0.10	0.12
Automatic: NIST	5.46	5.30	4.44	5.10

Results for WMT19:

- Best systems match humans in GCSE-like scoring.
- They score worse in pseudo-doc-aware DA.
- They are absolutely terrible on agreements.

... each technique provides a different picture.

Problems of Manual Evaluation

- Expensive in terms of time/money.
- Subjective (some judges are more careful/better at guessing).
- Not quite consistent judgments from different people.
- Not quite consistent judgments from a single person!
- Not reproducible (too easy to solve a task for the second time).
- Experiment design is critical!
- Black-box evaluation important for users/sponsors.
- Gray/Glass-box evaluation important for the developers.
- SRC-based allows to compare with humans.
- Sentence-level no longer relevant for large language pairs.

Automatic Evaluation

- Comparing MT output to reference translation.
- (Reference-less evaluation is called `QUALITY ESTIMATION`.)
- Fast and cheap.
- Deterministic, replicable.
- Allows automatic model optimization (“tuning”, MERT).
- Usually good for checking progress.
- Usually bad for comparing systems of different types.

BLEU (Papineni et al., 2002)

- Based on geometric mean of n -gram precision.
- $\approx$ ratio of 1- to 4-grams of hypothesis confirmed by a ref. translation

Src	The legislators hope that it will be approved in the next few days .	Confirmed
Ref	Zákonodárci doufají , že bude schválen v příštích několika dnech .	1 2 3 4
Moses	<u>Zákonodárci doufají , že bude schválen v</u> nejbližších <u>dnech</u> .	9 7 5 4
TectoMT	<u>Zákonodárci doufají , že bude</u> schváleno další páru volna .	6 4 3 2
Google	<u>Zákonodárci naději , že bude schválen v</u> několika příštích dnů .	9 4 3 2
PC Tr.	<u>Zákonodárci doufají že to bude schválený v</u> nejbližších <u>dnech</u> .	7 2 0 0

n-grams confirmed: none, unigram, bigram, trigram, fourgram

E.g. Moses produced 10 unigrams (9 confirmed), 9 bigrams (7 confirmed), ...

$$\text{BLEU} = \text{BP} \cdot \exp \left(\frac{1}{4} \log \left(\frac{9}{10} \right) + \frac{1}{4} \log \left(\frac{7}{9} \right) + \frac{1}{4} \log \left(\frac{5}{8} \right) + \frac{1}{4} \log \left(\frac{4}{7} \right) \right)$$

BP is “brevity penalty”; $\frac{1}{4}$ are uniform weights, the “denominator” equivalent for $\sqrt[4]{\cdot}$ in geometric mean in the log domain.

BLEU: Avoid Cheating/Gaming the Metric

- Confirmed counts “clipped” to avoid overgeneration.
- “Brevity penalty” applied to avoid too short output:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r \end{cases}$$

Ref 1: The cat is on the mat .

Ref 2: There is a cat on the mat .

Candidate: The the the the the the the .

⇒ Clipping: only $\frac{3}{8}$ unigrams confirmed.

Candidate: The the .

⇒ $\frac{3}{3}$ unigrams confirmed but the output is too short.

⇒ $\text{BP} = e^{1-7/3} = 0.26$ strikes.

The candidate length c and “effective” ref. length r calculated over the whole test set.

BLEU Properties

- Within the range 0-1, often written as 0 to 100%.
- Human translation against other humans: ~60%
- Google Chinese→English: ~30%, Arabic→English: ~50%.
- BLEU for individual sentences not reliable.
- More so with only 1 reference translation:

Src	" We ' ve made great progress .
Ref	" Učinili jsme velký pokrok .
Moses	" my jsme udělali <u>velký pokrok</u> .
TectoMT	" Udělali jsme velký pokrok .
Google	" My jsme dosáhli obrovského pokroku .
PC Translator	" udělali <u>jsme velký pokrok</u> .

Test Set Influence on BLEU

Havlíček (2007) evaluates the influence of:

- number of reference translations,
- translation direction.

on *human-produced* text (1 human translation against 4 others).

Refs	cs→en, Professionals				en→cs, Math Students			
	Indiv. Results			Avg	Indiv. Results			Avg
1	41.15	32.66	34.03	35.95	3.66	8.62	5.79	6.02
2	49.09	49.78	41.26	46.71	9.82	8.26	9.36	9.15
3	52.63			52.63	13.06			13.06

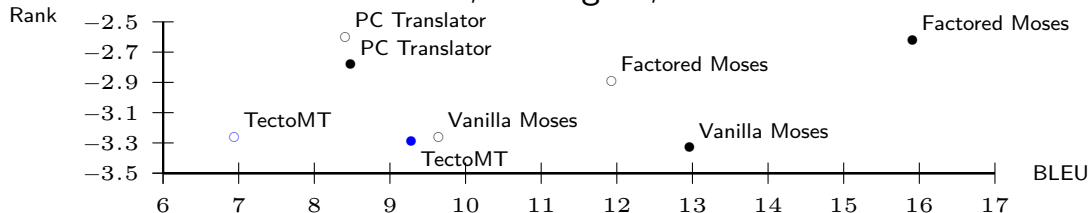
⇒ heavy dependence on the number of references.

More references allow to match more n-grams of MT output.

⇒ heavy dependence on the translation direction and quality.

Correlation with Human Judgments

BLEU scores vs. human rank, the higher, the better:



WMT08 Results	In-domain ●		Out-of-domain ○	
	BLEU	Rank	BLEU	Rank
Factored Moses	15.91	-2.62	11.93	-2.89
PC Translator	8.48	-2.78	8.41 !!	-2.60
TectoMT	9.28	-3.29	6.94	-3.26
Vanilla Moses	12.96	-3.33	9.64	-3.26

⇒ PC Translator nearly won Rank but nearly lost in BLEU.

Dirty Tricks

- PCEDT 1.0 (Čmejrek et al., 2004) contains test set with:
 - 1 English original,
 - 1 Czech translation,
 - 4 English back-translations (via Czech).
- Čmejrek et al. (2003) evaluate cs→en MT using all 5 English sentences: they include the original source among the references and report 5-fold average of BLEU (on 4 refs).
- The additional accepted variance in output increases BLEU compared to BLEU on the 4 back-translations only.

	5-fold Avg of 4-BLEU	4 refs only
PBT, no additional LM	34.8±1.3	32.5
PBT, bigger LM	36.4±1.3	34.2
PBT, more parallel texts, bigger LM	38.1±0.8	36.8

Improving BLEU in cs→en MT

A summary of old experiments. (Bojar et al., 2006; Bojar, 2006)

Deterministic pre- and post-processing	
similar tokenization of reference	+10.0 !!!
lemmatization for alignment	+2.0
handling numbers	+0.9
fixing clear BLEU errors	+0.5 !
dependency-based corpus expansion	+0.3
More parallel or target-side monolingual data	
out-of-domain parallel texts, bigger in-domain LM	+5.0
bigger in-domain LM	+1.7
out-of-domain parallel texts, also in LM	+0.4
adding a raw dictionary	+0.2

- Complicated methods bring a little.
- Data bring more.
- Huge jumps from superficial properties but just higher BLEU, same MT quality.

Finding Clear BLEU Losses

Missing bigram = all references contained it but not the hypothesis.

Superfluous bigram = the hypothesis contained it but none of the references.

Top missing bigrams:

19	, "	12	" said
12	of the	10	Free Europe
10	Radio Free	7	. "
6	L.J. Hooker	6	United States
6	in the	6	the United
6	the strike	...	

Top superfluous bigrams:

26	, ''	18	'' .
14	" said	12	, which
11	Svobodná Evropa	8	, when
8	the state	7	, who
7	J. Hooker	7	L. J.
7	company GM	...	

Four simple rules to improve BLEU by +0.2 to +0.5 on a particular test set:

'' .	→	. "		L. J. Hooker	→	L.J. Hooker
''	→	"		the U.S.	→	the United States

Technical Problems of BLEU

BLEU scores are not comparable:

- across languages.
- on different test sets.
- with different number of reference translations.
- with different implementations of the evaluation tool.
- There are different definitions of “reference length”:
Papineni et al. (2002) not specific. One can choose the shortest, longest, average, closest (the smaller or the larger!).
- Very sensitive to tokenization:
Beware esp. of malformed tokenization of Czech by foreign tools.

⇒ Use a fixed implementation, e.g. `sacreBLEU` (Post, 2018).

Fundamental Problems of BLEU

- BLEU overly sensitive to word forms and sequences of tokens.

Confirmed by Ref	Contains Error Flags	1-grams	2-grams	3-grams	4-grams
Yes	Yes	6.34%	1.58%	0.55%	0.29%
Yes	No	36.93%	13.68%	5.87%	2.69%
No	Yes	22.33%	41.83%	54.64%	63.88%
No	No	34.40%	42.91%	38.94%	33.14%
Total n -grams		35 531	33 891	32 251	30 611

30–40% of tokens not confirmed by reference but without errors.

⇒ Enough space for MT systems to differ unnoticed.

⇒ Low BLEU scores correlate even less.

Fixing Fundamental Issues of BLEU

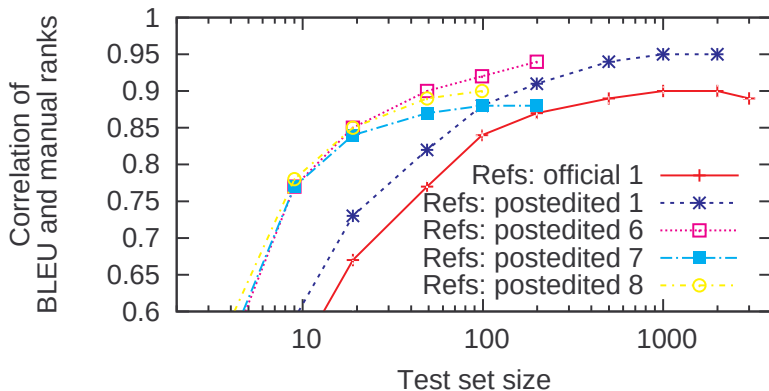
Evaluate coarser units:

- Lemmas or deep-lemmas instead of word forms:
 - e.g. SemPOS (Kos and Bojar, 2009): bags of t-lemmas.
- Sequences of characters:
 - e.g. chrF3 (Popović, 2015): F-score of character 6-grams.
- Use shorter or gappy sequences:
 - e.g. BEER (Stanojevic and Sima'an, 2014) uses characters and also pairs of (not necessarily adjacent) words.

Use better references:

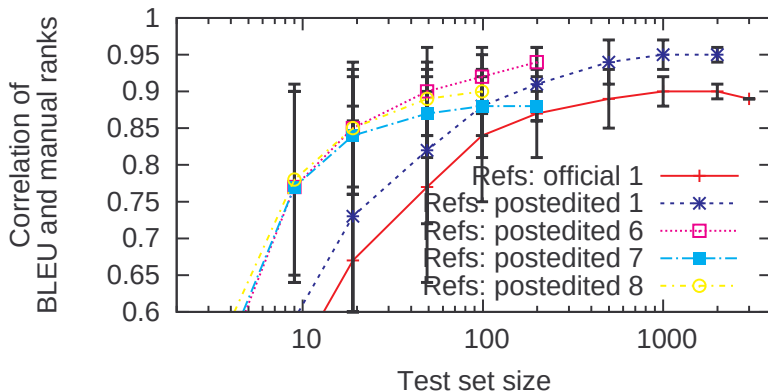
- Using more references alone helps.
- Post-edited references serve better.
 - e.g. HTER (Snover et al., 2006): Measuring edit distance to manually corrected output.

Post-Edited References Serve Better



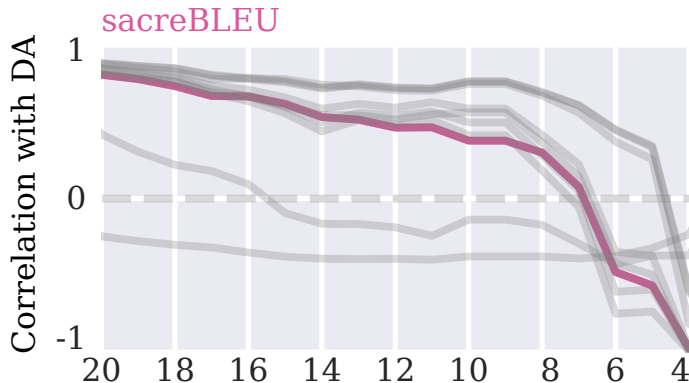
- Refs created by post-editing serve better than independent ones.
- 100 sents with 6–7 postedited refs as good as 3k indep refs.

Post-Edited Refs Better



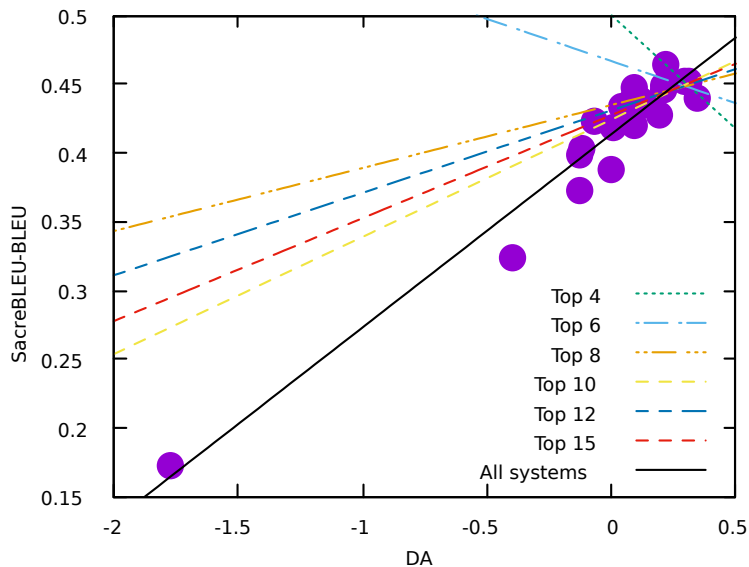
- ... but error bars quite wide
⇒ specific sentences important.

Fundamental Problem of Correlation



- Correlation depends on the *underlying set of MT systems*.
- Often poor correlation when only top-scoring systems are considered, see Ma et al. (2019).

Fundamental Problem of Correlation



“LLMs as Judges”

- We delegate the complex task to LLMs, hoping in their reliability.

In the following:

- Zdeněk Kasner's experiments with automatic error span annotation.
- Pavel Šindelář's experiments with evaluation of answers (Sensemaking 2025).

Experience with English-**{Maltese, Spanish}** translation

- Errors: omissions, negations, incorrect gender, superfluous words
- Iterative prompt updates:
 - a. do not translate the source text from scratch;
 - b. review the initial translated text;
 - c. indicate the category of the error;
 - d. do not indicate errors where none are found in the text,
- With the initial prompt, the LLM **tried too hard** to find errors → false positives
- Fixed by “most of the translations are correct” → lower recall
- Temperature seemed to impact consistency of responses

Experiments: Our data

- 10 DGT documents that were **fully annotated** for errors
- EN→CS translations, various domains

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Budget
Revenue and Multiannual financial
framework
Revenue inspections, control and
enforcement
Brussels, 16 December 2021
Pre-inspection questionnaire
INSPECTION OF MEMBER STATES' SYSTEM
FOR COLLECTING ANTI-DUMPING AND
COUNTERVAILING DUTIES
Why this questionnaire?
DG BUDG has selected the control strategy
for anti-dumping and countervailing duties
as one of the topics for traditional own
resources (TOR) inspections in 2022.

cs-tra

eval-feedback-xlsx-ann0

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2022.

gpt-oss:120b: open reasoning model from OpenAI

↓ 2.7M Downloads ⌚ Updated 4 days ago

OpenAI's open-weight models designed for powerful reasoning, agentic tasks, and versatile developer use cases.

tools

thinking

cloud

20b

120b

<https://ollama.com/library/gpt-oss:120b>

Experiments: Our prompt

Your task is to **identify errors in the translation** and classify them.

(...)

Output the errors as a **JSON object** with a single key "annotations".

(...)

Accuracy errors address the correspondence of content between the source and the target text. Accuracy errors inappropriately change the intended meaning of the source text.

(...)

A minor error: a) does not seriously affect the usability, understandability, or reliability of the content (even if correction is still needed).

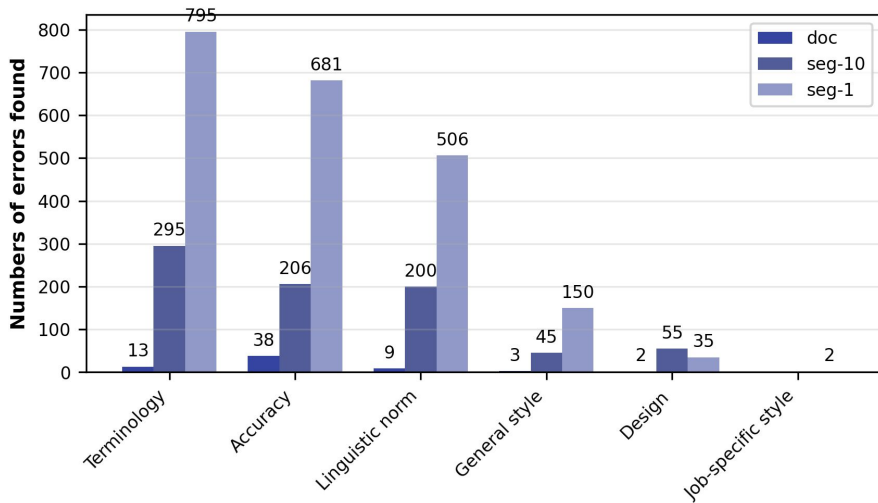
(...)

If there is **nothing to annotate in the text**, "annotations" will be an empty list.

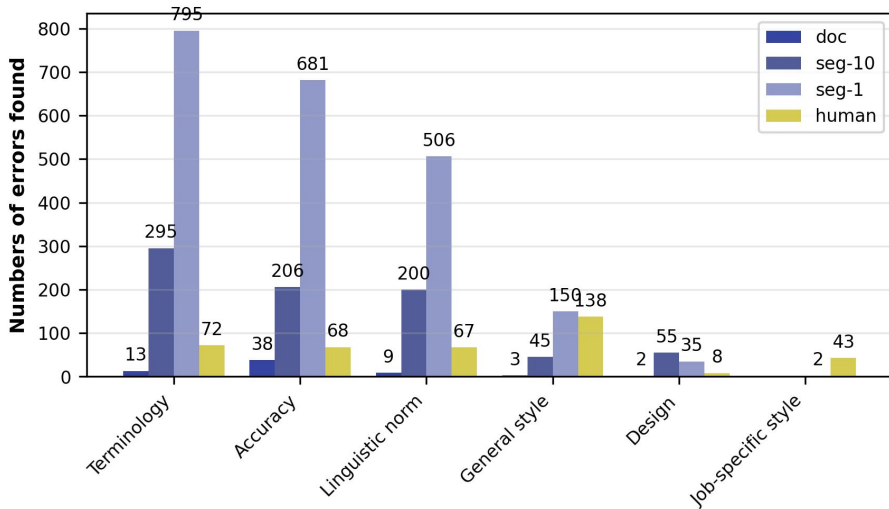
accy-		Accuracy (Minor)
accy+		Accuracy (Major)
term-		Terminology (Minor)
term+		Terminology (Major)
lnorm-		Linguistic norm (Minor)
lnorm+		Linguistic norm (Major)
stjob-		Job-specific style (Minor)
stjob+		Job-specific style (Major)
stgen-		General style (Minor)
stgen+		General style (Major)
dsgn-		Design (Minor)
dsgn+		Design (Major)
unknown		Unknown error category

2832 text segments annotated with
gpt-oss-120b overnight 💪

Document chunking matters... A LOT



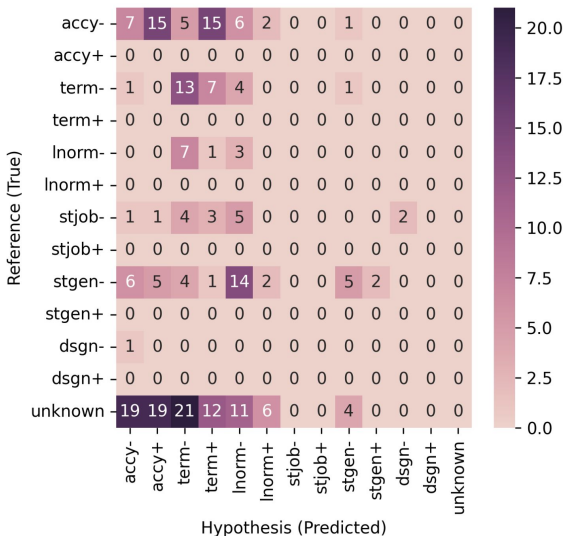
Document chunking matters... A LOT



More statistics w.r.t. document size

Metric	doc	seg-10	seg-1	human
Total examples considered	8	260	2564	260
Examples with annotations	8	244	1421	126
Examples without annotations	0	16	1143	134
Percentage of empty examples	0.0%	6.15%	44.58%	51.54%
Total annotations	65	801	2169	688
Annotations per example (avg)	8.12	3.08	0.85	2.65
Average annotation length (chars)	26.06	21.05	16.53	8.05

Did the error categories match?



Is there any correlation on seg-10?

Hard matching on seg-10

(→ *the categories have to match*)

Precision: 0.009

Recall: 0.017

F1: 0.012



Soft matching on seg-10

(→ *categories are irrelevant*)

Precision: 0.069

Recall: 0.128

F1: 0.09



Case study: budg-2021, seg-10

source

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human

cs-tra  eval-feedback-xlsx-ann0 

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model

cs-tra  gpt-oss-120b-ann0 

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Case study: budg-2021, seg-10

source

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model ✗

cs-tra

gpt-oss-1:

accy- (The modifier "příjmů"
(Revenue) is only attached
to the last item, leaving
"Inspekce" and "kontrola"
without the intended
qualifier.)
logo_ec_17_colors_300dpi
EVROPSKÁ KOMISE
Rozpočet
Příjmy a výdaje v finanční rámc

Inspekce, kontrola a vymáhání příjmů

V Bruselu dne 16. prosince 2021

Case study: budg-2021, seg-10

source

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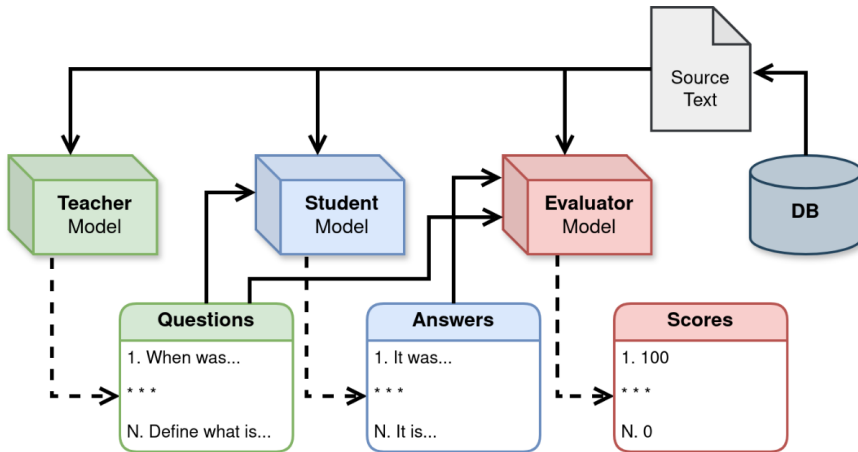
model

U ČLENSKÝCH STÁTŮ PRO VÝBĚR
H A VÝROBY (The abbreviation
“(TOR)” present in the
source is omitted in the
translation, reducing
precision.)
a vyrovnávací o jedno z
oly tradičních vlastních zdrojů

Models Overview

- Smaller models are probably useless, larger/more recent models = better.
- Reasoning models give more interesting outputs.
- Models it makes sense to try:
 - GPT-OSS (open source, can be run on premises, will need tweaking)
 - GPT 4o (large commercial model)

Sensemaking 2025



- In 2026, we run the task again, focusing on **Evaluators** only.

Rating adversarial items

rating by	expert answer	non adversarial	adversarial answer	nonsense answer	material swap
expected	100		10 – 20	0	0
Gemma-3(27b)	95±16	83±27	67±40	33±40	59±43
Mistral(7b)	73±47	62±48	55±50	59±49	58±49
baseline	81±35	66±37	49±43	13±27	26±36

- **expert answers:** Original question and reference answer.
- **non-advers.:** Original question and regular answer by a Student system.
- **advers. answer:** Question kept, answer taken from another dataset part.
- **nonsens. answer:** Question kept, answer words shuffled or replaced by newspaper text.
- **material swap:** Question and answer swapped into another dataset part.

Note also the big confidence intervals.

Rating adversarial items

Some decrease but not big enough.

rating by	expert answer	non adversarial	adversarial answer	nonsense answer	material swap
expected	100		10 – 20	0	0
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- **expert answers:** Original question and reference answer.
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- **advers. answer:** Question kept, answer taken from another dataset part.
- **nonsens. answer:** Question kept, answer words shuffled or replaced by newspaper text.
- **material swap:** Question and answer swapped into another dataset part.

Note also the big confidence intervals.

Rating adversarial items

Large decrease when answer nonsensical.

rating by	expert answer	non adversarial	adversarial answer	nonsense answer	material swap
expected	100		10 – 20	0	0
Gemma-3(27b)	95±16	83±27	67±40	33±40	59±43
Mistral(7b)	73±47	62±48	55±50	59±49	58±49
baseline	81±35	66±37	49±43	13±27	26±36

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Note also the big confidence intervals.

Rating advice

Small decrease when both swapped.

rating by	expert answer	non adversarial	adversarial answer	nonsense answer	material swap
expected	100		10 – 20	0	0
Gemma-3(27b)	95±16	83±27	67±40	33±40	59±43
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Note also the big confidence intervals.

Empirical Confidence Intervals

In statistics, confidence intervals indicate how well was a parameter (e.g. the mean) of a random variable with *known/assumed distribution* estimated from a set of repeated measurements.

- We don't want to assume any distribution!
- How to “repeat” experiments with a deterministic MT system?

Use “bootstrapping” (Koehn, 2004):

1. Obtain 1000 different test sets:
Randomly select sents., repeat some, ignore some, preserving test set size.
2. Sort by the score.
3. Drop top and bottom 2.5% (i.e. 25 out of 1000) results.

⇒ The lowest and highest remaining scores are 95% empirical confidence interval around the score obtained on the full test set.

End-to-end vs. Component Eval.

- Similar to black vs. glass box evaluation and translation vs. task-based evaluation.
- Evaluation of a single component may not correlate with overall performance of the system.

Pre-processing	Symmetrization	Alignment Error Rate	BLEU
Lemmas + singletons	Intersection	14.6	30.8
Lemmas	Intersection	15.0	29.8
Lemmas	Union	17.2	32.0
Lemmas + singletons	Union	17.4	31.9
Baseline (word forms)	Union	25.5	29.8
Baseline (word forms)	Intersection	27.4	28.2

Data by Bojar et al. (2006). See also e.g. Lopez and Resnik (2006).

Summary, The Moral of the Story

Metrics drive research:

- Measure the property that “saves money” in your application.
- Design automatic metrics to correlate with humans.
 - Very carefully with “LLMs as judges”.

Comparisons of automatic scores trustworthy

only under all the following:

- a single test set was used (of your domain of interest),
- evaluated by a single evaluation tool (hopefully without bugs),
E.g. for BLEU different tools tokenize and define ref. length differently.
- the metric reflects your final objective (AER vs. BLEU),
- confidence intervals are estimated.

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